Optimal Carbon Tax with Dynamic Skill Heterogeneity

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July 13, 2026†
Work in Progress: Latest version

Abstract

How should a government price carbon when it cares about redistribution and emissions? I extend the dynamic Mirrlees model to two goods, one polluting, with privately observed skills that evolve over the life cycle. Under weakly separable preferences, the optimal carbon wedge equals the social cost of carbon while the labor and savings wedges keep their standard forms. The below-Pigou prescription of the second-best tradition disappears once the income tax is set optimally. Because real carbon taxes are linear, I also solve the problem when the instrument must be uniform and find that the restriction constrains redistribution but not externality correction. In a US calibration, the optimal energy wedge rises with skill, so the regressivity of real carbon taxes comes from their linearity. However, a flat tax at the social cost of carbon captures 98% of the welfare gain of the full optimum, so the income tax should focus on redistribution and the carbon price on externality correction.

Keywords: *Carbon tax, dynamic Mirrlees taxation, second-best environmental policy*
JEL Classification: *D62; D82; H21; H23*

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†I thank my advisor Guillaume Sublet for his invaluable guidance and support, as well as the members of his workshop and the participants of the CIREQ Weekly PhD Student Research Workshop, the McGill University microbreakfast, the University of Montréal macro brownbag, the CIREQ-CIRANO-RRECQ Environmental and Natural Resource Economics Workshop, the 19th CIREQ PhD Students' Conference, the 59th CEA Annual Conference, the 35th CREEA Annual Conference, the 65th SCSE Conference, the 60th CEA Annual Conference, the 25th LAGV conference, and the 7th WCERE conference for their comments and feedback. I am grateful for the support and funding from the Fonds de Recherche du Québec Société et Culture (FRQSC) through the Doctoral Training Scholarships program. DOI: https://doi.org/10.69777/329014.

1 Introduction

Carbon pricing is the economist's favorite climate policy and the electorate's least favorite tax. France froze its carbon tax after the *gilets jaunes* protests; Canada repealed its consumer carbon tax in 2025. The charge is the same everywhere: the tax unfairly burdens low-income households. The tools of carbon policy cannot even weigh that charge, because the integrated assessment models behind official carbon prices (Environment and Climate Change Canada (2023), Interagency Working Group on Social Cost of Greenhouse Gases (2021)) treat the world, or each country, as a single representative agent, with no rich or poor households to compare.

Whether carbon taxes are in fact regressive has been contested for three decades, and the contest turns on the life cycle. When households are ranked by annual income, taxes on gasoline and other energy fall far more heavily on the poor; ranked by lifetime or permanent income, the burden is much flatter, because households smooth consumption and today's low earners include the young and the temporarily unlucky (Poterba 1989; Hassett et al. 2009). The welfare-relevant burden of a carbon tax is a lifetime object, not an annual one. A model that speaks to the regressivity debate on its own terms must therefore be a lifecycle model.

So far, though, the lifecycle perspective has stayed within positive economics: it measures the incidence of existing carbon taxes without asking how the tax should be designed in the first place. Barrage (2019) and Douenne et al. (2026) are the exceptions, but the first ties the tax system to an *ad hoc* revenue target rather than optimal redistribution, and the second restricts attention to linear instruments, with no inverse-Euler savings margin. The design question is open. How should a government price carbon when it cares about redistribution, and does pollution change how income should be taxed?

I answer the question in a dynamic Mirrlees economy with two goods, one clean and one dirty, in which agents draw privately observed skills over the life cycle and their consumption of the dirty good feeds a stock externality. To my knowledge this is the first dynamic-Mirrleesian derivation of the optimal carbon tax, with private information, nonlinear history-dependent taxes, and a positive savings wedge. Relative to the static irrelevance theorem of Kaplow (2012) and Jacobs and de Mooij (2015), the dynamic economy delivers three things the static case cannot: a savings margin and its inverse-Euler wedge, an age profile and history dependence in the carbon wedge, and the welfare cost of restricting the carbon tax to an anonymous uniform price. All three are quantitative objects, and I measure them in a calibration to US data.

The central finding is a *separation.* Under weak separability, the carbon wedge equals the social cost of carbon, the labor wedge keeps the Golosov et al. (2016) ABCD form, and the savings wedge keeps its inverse-Euler form. Three margins do three jobs without interacting. Redistributing through the optimal nonlinear income tax restores the split between corrective and redistributive taxation that the Ramsey second best breaks: the below-Pigou prescription of that tradition was never a property of distortionary taxation as such, only of its linear instruments, and it disappears once the income tax is set optimally under private information. The carbon price should not be bent for lifetime distribution; the nonlinear income tax and the climate dividend do that work.

I then quantify the separation by solving the optimal mechanism globally. The optimal wedge on energy is the social cost of carbon scaled up by a history-dependent Corlett-Hague premium: it starts at the Pigouvian level, rises with age as the labor wedge does, and ends the working life about 9% above the Pigouvian price. That premium is small in welfare. A flat carbon tax at the social cost of carbon captures 98% of the welfare gain of the full optimum, and the wedge's entire Mirrleesian fine structure (its age profile, history dependence, and progressivity in skill) is worth a few thousandths of a percent of consumption. This near-sufficiency is the separation in welfare terms: pricing carbon at the social cost of carbon and redistributing through the income tax is close to optimal even when preferences are nonseparable and energy is a leisure complement.

The deviations from the Pigouvian benchmark that do arise are small, signed by how each good complements labor, and progressive in skill. Late in the working life the optimal energy wedge reaches about 70% for the top skill quintile against 57% for the bottom, both above the Pigouvian level near 50%: the optimal carbon wedge rises with skill rather than falling. The regressivity of *real* carbon taxes is therefore a property of flat instruments and of energy's status as a necessity, not of the optimum. This finding is the normative counterpart the positive incidence debate never supplied, and it complements Douenne et al. (2026) and Douenne et al. (2025), who reach a near-Pigouvian conclusion from the Ramsey side; here I derive the preference conditions that make the separation exact and the channels that break it.

Section 2 places the argument in the literature. Section 3 sets up the two-good dynamic Mirrlees economy with a stock externality. Section 4 derives the optimal commodity wedge in a dynamic Mirrlees economy without an externality, isolating the single margin whose vanishing is the separation. Section 5 introduces the externality and states the separation and the modified social cost of carbon. Section 6 restricts the commodity instrument to a uniform anonymous price, the form real carbon taxes take, and prices the welfare cost of the restriction. Section 7 reverses the exercise, freezing the labor, savings, and commodity wedges in turn at exogenous schedules and re-deriving the remaining wedges against them; the separation fails there in signable directions. Section 8 solves the mechanism numerically, calibrated to US data, and measures both the size of the optimal wedge and the welfare cost of simpler carbon-pricing rules. Section 9 concludes.

2 Literature review

This paper sits between two literatures that reach opposite verdicts on how distortionary taxation should bend the carbon price, and draws on a third that has the tools to adjudicate but has never been aimed at the environment. The disagreement is consequential. Should a government facing a distortionary tax system price carbon below marginal damages, because the environmental tax compounds distortions elsewhere, or does that markdown dissolve once the income tax is itself set optimally? The answer decides whether tight fiscal constraints call for timid or full carbon pricing, and it turns on how flexible the redistribution instrument is.

The Ramsey tradition answers that the markdown is real and grows with the distortion. Sandmo (1975) shows that with linear instruments the optimal pollution tax is additive, marginal damages entering the polluting good's formula alone, but divided by the marginal cost of public funds, which exceeds one under distortionary labor taxation: the optimal pollution tax then lies strictly below marginal damages. Bovenberg and de Mooij (1994) trace the gap to a tax-interaction effect, by which the environmental tax amplifies the pre-existing labor-supply distortion, and Parry et al. (1999) place the optimal rate at 63 to 78% of marginal damage. Carried into the dynamic climate-economy models that discipline policy, the verdict survives. Golosov et al. (2014) supply the closed-form representative-agent benchmark, and Barrage (2019) puts the optimal dynamic carbon tax 8 to 24% below it, driven by the same public-funds division and by rigid capital-tax instruments. This whole arc, though, is representative-agent or observable-type: inequality enters neither the objective nor the incentive constraints.

Adding heterogeneous agents to this dynamic Ramsey frontier softens the verdict, and the papers that do so are this paper's closest antecedents. Douenne et al. (2026) solve a Ramsey problem with heterogeneous agents, a calibrated climate module, and linear instruments, and find the optimal carbon tax essentially Pigouvian, inequality lowering it by only about 4%; Douenne et al. (2025) show the result is robust to uninsurable income risk and tight fiscal constraints. Both Barrage (2019) and Douenne et al. (2026) also derive third-best carbon taxes with an income-tax rate frozen at its observed level, an exercise Section 7 revisits with Mirrleesian instruments. The distributional stakes are real: Känzig (2023) finds carbon-pricing shocks fall roughly three times harder on low-income than on high-income households, and the dispersion of energy spending *within* income groups exceeds the dispersion *across* them (Cronin et al. 2019; Pizer and Sexton 2019), precisely the heterogeneity a type-specific wedge would respond to.

Restricting the instrument menu can even reverse the near-Pigouvian verdict: van der Ploeg et al. (2025) show that a uniform carbon tax paired only with a uniform dividend can optimally exceed marginal damages, and Hänsel et al. (2022) that a mandated uniform carbon tax makes the optimal income tax more progressive. A parallel policy debate presses for carbon prices that differ across individuals and over their cumulative choices (Chancel 2022; Dietsch 2024), a demand that no incentive-compatible optimal-tax analysis has yet grounded. All of this heterogeneous-agent work, though, stays within the Ramsey paradigm, with linear instruments, observable types, and period-by-period planning: none of it carries a nonlinear income tax or an inverse-Euler savings wedge.

A parallel static Mirrleesian literature argues the opposite: that the Ramsey markdown is not a robust feature of distortionary taxation but an artifact of the linear-instrument restriction. Cremer et al. (1998) show that when individual consumption is observable and the nonlinear income tax is optimized, Sandmo additivity survives and the public-funds division disappears, and Kaplow (2012) makes the argument cleanest: any environmental reform can be rendered distribution-neutral by adjusting the income tax type by type, so that a move to the Pigouvian rate is Pareto-improving. Jacobs and de Mooij (2015) formalize it: an optimized nonlinear income tax sets the marginal cost of public funds to one, and when goods and environmental quality are weakly separable from labor the optimal pollution tax equals marginal damages. When separability fails the tax deviates in signable directions: above Pigouvian for a leisure complement, the Corlett-Hague case substantiated for gasoline by West and Williams III (2007), below for a necessity with non-homothetic Engel curves (Jacobs and van der Ploeg 2019), and either way once the redistributive role of commodity taxation is in play (Christiansen 1984). The calibrated two-good static-Mirrlees treatment nearest to this paper's setting is Cremer et al. (2010). All of this is static, however: it has the nonlinear income tax but no life cycle, no savings, and no age profile of earnings, so it cannot discipline the time path of the carbon tax that integrated assessment models require.

The tools for a dynamic treatment come from a third literature, which built the theory of optimal taxation under private information but abstracted from externalities. Golosov et al. (2003) establish two results this paper starts from. The inverse Euler equation makes the optimal savings wedge strictly positive, because private saving makes future incentive constraints harder to satisfy, overturning the Chamley-Judd zero-capital-tax prescription. And under weak separability, Atkinson-Stiglitz uniform commodity taxation remains optimal within each period. Kapička (2013) extends the first-order approach to persistent shocks, Farhi and Werning (2013) characterize the labor wedge over the life cycle, and Golosov et al. (2016) give its tightest form, the ABCD decomposition this paper builds on. The same apparatus has already been turned on lifetime-varying taxes without reaching the environment, which makes the gap conspicuous: Weinzierl (2011) finds large welfare gains from conditioning the income tax on age alone. The nearest neighbor with an externality, Chen and Ren (2024), embeds a generic consumption externality in a two-period economy but has neither a genuine cross-agent pollutant with output and utility damages nor an explicit clean/dirty commodity wedge, and the closest environmental hybrid, Crampes and Ladoux (2024), prices energy efficiency in a two-period Mirrleesian setting without persistent skill shocks or the inverse-Euler structure.

These three literatures leave the same gap: each supplies part of what a lifecycle optimal carbon tax needs, and none supplies all of it. This paper fills that gap, and its three contributions map onto the three strands. It carries the dynamic Mirrleesian apparatus from one consumption good to two; it extends the Jacobs and de Mooij (2015) irrelevance theorem to a dynamic economy with private information and, under nonseparability, signs the deviations Ramsey treatments cannot recover; and it returns the carbon price to the anonymous linear form real policy uses and prices the cost of that restriction. The result complements Douenne et al. (2026) rather than competing with it: where they measure, quantitatively and from the Ramsey side, that the optimal carbon tax is near-Pigouvian, this paper derives, analytically and from the Mirrleesian side, the preference conditions under which corrective and redistributive taxation separate exactly, and the channels that break the separation.

3 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2003), Farhi and Werning (2013), and Golosov et al. (2016). The two goods are a clean numéraire and a polluting good. To isolate how household income inequality shapes the optimal carbon tax, I abstract from international and intergenerational equity, and consider a single country and a single generation.

The economy lasts $T+1$ periods. Each household consumes a clean good c_t and a polluting good q_t, supplies labor l_t, and is exposed to the aggregate pollution stock S_t. Expected lifetime utility is

$$\mathbb{E}_0\left[\sum_{t=0}^{T}\beta^t U(c_t, q_t, l_t, S_t)\right]$$

$\beta \in (0,1)$ is the discount factor and the clean good c is the numéraire, whose price is normalized to 1. A unit of output transforms one-for-one into either good, so the producer price of the dirty good is also 1. Utility is time-separable across periods but may be nonseparable across its arguments. Consuming the dirty good generates emissions that accumulate into a persistent pollutant stock S_t, a function of current and past aggregate emissions,

$$S_t = S\left(Q_0, Q_1, ..., Q_t\right),$$

where $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta]\, dF_0(\theta)$ is aggregate consumption of the dirty good at date t. This is the carbon-cycle form of Golosov et al. (2014) and Joos et al. (2013): each past emission flow leaves its own trace in the current stock. The leading example keeps a fraction δ of the stock from one period to the next,

$$S_t = \sum_{j=0}^{t} \delta^j Q_{t-j},$$

equivalently the recursion $S_t = \delta S_{t-1} + Q_t$ from a zero initial stock. I leave the accumulation function otherwise unrestricted. To keep the state finite, I assume emissions stop affecting the stock after s periods,

$$S_t = S\left(Q_{t-s}, ..., Q_t\right),$$

so the pass-through of current emissions, $\partial S_{t+j}/\partial Q_t$, vanishes for $j > s$ by construction. Under geometric decay the truncation drops the tail δ^j for $j > s$, a good approximation once δ^s is negligible.

Aside from the polluting good, the model follows the basic setup of the dynamic Mirrlees literature. At $t = 0$, agents draw an initial skill (type) θ_0 from a distribution $F_0(\theta)$. Later skills follow a Markov process $F_t(\theta|\theta_{t-1})$. At period t, each agent knows their skill draws up to time t, which I denote by a skill history vector $\theta^t = (\theta_0, ..., \theta_t)$. Θ^t is the set of all possible skill histories.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Everyone observes total output, but skill θ_t and labor supply l_t are private. Consumption of both goods is also publicly observable[1]. Let $c_t(\theta^t), q_t(\theta^t) : \Theta^t \to \mathbb{R}_+$ and $y_t(\theta^t) : \Theta^t \to \mathbb{R}_+$ be the agent's allocation of the clean good, the polluting good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \to \Theta^t$ describe how an agent with skill history θ^t reports it, and let Σ^t be the set of all possible reporting strategies. Capital transfers resources across periods at gross return $R > 0$. The social welfare function is

$$\int_0^\infty \alpha(\theta)\, \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a non-negative Pareto weight on the welfare of an agent with initial skill θ, normalized so that $\int_0^\infty \alpha(\theta)\, dF_0(\theta) = 1$. If the planner is utilitarian, $\alpha(\theta) = 1$ for all θ. Let U_c and U_q denote the partial derivatives with respect to the clean and polluting goods, and U_l with respect

[1] This is a plausible assumption for some goods such as electricity and air travel. For goods whose consumption cannot be observed, the planner is further constrained and the unrestricted wedges of Sections 4 and 5 no longer apply. The uniform tax of Section 6 survives: an anonymous linear price requires observing total expenditure, not its composition.

to labor. Since $y = \theta l$, I can write $U(c, q, l, S)$ as $U\left(c, q, \frac{y}{\theta}, S\right)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ.

Assumption 3.1 (preferences)**.**
U is twice continuously differentiable with $U_c, U_q > 0$, $U_l, U_S < 0$, $U_{cc} \leq 0$, $U_{qq} \leq 0$, $U_{ll} \leq 0$, jointly concave in (c, q, l), and satisfies Inada conditions in (c, q). Single crossing holds:

$$\frac{\partial}{\partial \theta} \frac{U_l(c, q, y/\theta)}{U_c(c, q, y/\theta)} \geq 0 \qquad \text{for all } (c, q, y, \theta).$$

Assumption 3.2 (shock process)**.**
For all $t \geq 1$, f_t is differentiable in both arguments with $f_t' = \partial f_t / \partial \theta$ and $f_{2,t} = \partial f_t / \partial \theta_{t-1}$. For all θ_{t-1}, the ratio

$$\frac{\theta_{t-1} \int_\theta^\infty f_{2,t}(x|\theta_{t-1})\, dx}{\theta f_t(\theta|\theta_{t-1})}$$

is bounded, and $\lim_{\theta \to \infty} \frac{1 - F_t(\theta|\theta_{t-1})}{\theta f_t(\theta|\theta_{t-1})}$ is finite.

Any admissible allocation must be incentive compatible: the agent must weakly prefer truthful reporting to any other strategy. The stock S_t is an aggregate, so no single agent's deviation moves it, and it enters both sides of the constraint identically. For every initial type θ and every reporting strategy,

$$\begin{aligned} &\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ &\qquad \geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U\left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned}$$

Over a finite horizon, global incentive compatibility is equivalent to the absence of a profitable one-shot deviation after every history (Fernandes and Phelan 2000/04/01/; Kapička 2013). I follow the first-order approach of Kapička (2013), Farhi and Werning (2013), and Golosov et al. (2016), replacing these constraints with their local counterpart. Let $u_t(\theta^t)$ be the lifetime utility an agent obtains from reporting truthfully at every date. Truth-telling is locally optimal when the envelope condition holds.

$$\dot{u}_t(\theta^t) = U_\theta\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta\, w_{2,t}(\theta^t)$$

$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta_{t+1})\, \partial_{\theta_t} f_{t+1}(\theta_{t+1}|\theta_t)\, d\theta_{t+1}$ is the marginal continuation value of type θ^t, and the dot denotes a derivative with respect to the current draw θ_t. The continuation term is the future information rent of the current draw: with i.i.d. shocks, $\partial_{\theta_t} f_{t+1} = 0$ and it vanishes, since today's skill then carries no information about tomorrow's. Under Assumptions 3.1 and 3.2, this local condition is necessary for incentive compatibility (Kapička 2013); monotonicity requirements on the allocation belong to the sufficiency direction, not to necessity. The local conditions define a relaxed problem: they are necessary but not sufficient for global incentive compatibility, and no general primitive restriction guarantees sufficiency in dynamic models with multiple goods (Golosov et al. 2016). I solve the relaxed problem throughout and verify global incentive compatibility numerically in Section 8, following Farhi and Werning (2013).

The aggregate resource constraint must hold in every period,

$$\int_0^\infty \mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t)|\theta]\, dF_0(\theta) + K_{t+1} \leq \int_0^\infty \mathbb{E}_0[y_t(\theta^t)|\theta]\, dF_0(\theta) + RK_t,$$

where K_t is the aggregate capital stock, with K_0 given and $K_{T+1} = 0$. Dividing each period's constraint by R^t, summing over t, and using these boundary conditions to telescope the capital terms consolidates the sequence into a single intertemporal constraint,

$$\int_0^\infty \mathbb{E}_0\left[\sum_{t=0}^{T} \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t}\middle|\theta\right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0\left[\sum_{t=0}^{T} \frac{y_t(\theta^t)}{R^t}\middle|\theta\right] dF_0(\theta) + RK_0.$$

The planner chooses allocations to maximize welfare subject to the local incentive constraint, the intertemporal resource constraint, and the pollutant law of motion.

$$\max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t,\ t=0,\ldots,T}} \int_0^\infty \alpha(\theta) \left(\mathbb{E}_0\left[\sum_{t=0}^{T} \beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)\middle|\theta\right]\right) dF_0(\theta)$$

Three wedges, the implicit marginal taxes on the labor, savings, and commodity margins, describe the distortions the optimal allocation imposes.

$$1-\tau_t^y(\theta^t) \equiv -\frac{U_l\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\theta_t\, U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}$$

$$1-\tau_t^s(\theta^t) \equiv \frac{1}{\beta R}\,\frac{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\mathbb{E}_t\left[U_c\left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}, S_{t+1}\right)\right]}$$

$$1+\tau_t^q(\theta^t) \equiv \frac{U_q\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}$$

Table 1 collects six preference statistics used throughout.

Table 1: Preference elasticities and complementarities

Symbol	Definition	Interpretation
$\epsilon_t(\theta)$	$\frac{U_{ll,t}\, l_t}{U_{l,t}}$	Inverse Frisch elasticity of labor supply
$\sigma_t(\theta)$	$-\frac{U_{cc,t}\, c_t}{U_{c,t}}$	Inverse elasticity of intertemporal substitution of the numéraire
$\sigma_{cq,t}(\theta)$	$-\frac{U_{cq,t}\, q_t}{U_{c,t}}$	Cross-curvature of the numéraire's marginal utility
$\gamma_{c,t}(\theta)$	$\frac{U_{cl,t}\, l_t}{U_{c,t}}$	Complementarity of the clean good with labor
$\gamma_{q,t}(\theta)$	$\frac{U_{ql,t}\, l_t}{U_{q,t}}$	Complementarity of the polluting good with labor
$\xi_t(\theta)$	$\frac{U_{Sl,t}\, l_t}{U_{S,t}}$	Complementarity of the pollutant with labor

Most of the commodity-wedge results hinge on the gap $\gamma_{c,t} - \gamma_{q,t}$ between the clean and dirty goods' complementarities with labor. Each γ measures how a good's marginal utility moves with labor supply. A good with $\gamma > 0$ is a labor complement, wanted more by an agent who works more; a good with $\gamma < 0$ is a leisure complement, wanted more by an agent who works less. The gap asks which of the two goods leans more toward leisure. When it is positive, as it is for energy, the dirty good is the relative leisure complement, and an agent who under-reports skill and takes the extra leisure spends more of the marginal dollar on it. A tax on that good falls hardest on exactly the mimicker the planner screens against, so it relaxes an incentive constraint the income tax alone leaves binding. When the gap closes, the two goods co-move with labor identically, taxing one rather than the other does nothing for screening, and the commodity wedge is zero. This is the dynamic counterpart of the Atkinson-Stiglitz result.

The same logic reaches the pollution stock, through ξ_t. It records how the marginal damage of the stock moves with labor: $\xi_t > 0$ when an agent who works more is hurt more by an extra unit of the stock, so its damage, like that of a labor-complementary good, falls hardest on the hardest workers. For the carbon wedge what matters is the gap with the clean good, $\xi_t - \gamma_{c,t}$. When the two coincide, abatement changes the high type's payoff from truth-telling and from mimicking by the same amount, leaves the screening problem untouched, and the carbon price is the pure Pigouvian social cost of carbon. When pollution bites harder on workers, $\xi_t > \gamma_{c,t}$, abatement rewards the truthful high type, who works, over the mimicker, who under-reports and takes leisure; it slackens the binding incentive constraint and doubles as a screening device, so the planner prices carbon above the Pigouvian level. When $\xi_t < \gamma_{c,t}$ the sign flips and the price falls below. Section 5 makes this precise.

4 The commodity wedge without an externality

The second good adds exactly one new margin, the commodity wedge, whose sign is set by the gap $\gamma_{c,t} - \gamma_{q,t}$ introduced earlier. I characterize all three wedges in a clean two-good economy, before pollution enters, as the baseline for the results in Section 5. The problem is recursive, following Kapička (2013)[2].

$$V_t(\hat{w}, \hat{w}_2, \theta_-) = \min_{c,q,y,u,w,w_2} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta) \right] f_t(\theta|\theta_-) d\theta$$

$$\text{s.t.}$$

$$\dot{u}(\theta) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta|\theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta$$

$$u(\theta) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w(\theta)$$

[2]Kapicka's consumption set is an arbitrary convex and compact subset of $\mathbb{R}^N$, so consumption need not be a scalar and can be a vector of multiple goods.

Here $\bar{\alpha}_t(\theta)$ is the continuation Pareto weight attached to delivering utility to type θ. It equals the primitive Pareto weight in the initial period, $\bar{\alpha}_0(\theta) = \alpha(\theta)$, and $\bar{\alpha}_t(\theta) = 1$ for $t > 0$. As in Golosov et al. (2016), the lifetime Pareto weights enter only at date 0; from date 1 on, efficiency weights each type by its marginal utility of the numéraire through the promise-keeping constraint, so the continuation weight collapses to one. Solving the problem yields the three wedges below, the labor and savings wedges written in the marginal utility of the numéraire.

Result 4.1 (Optimal wedges).
The labor wedge is

$$\frac{\tau^y}{1-\tau^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,D_t(\theta)$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta)$$

$$B_t(\theta) = \frac{1-F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right](1-\lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x))\,\frac{f_t(x)dx}{1-F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}}\,\frac{U_{c,t}(\theta)}{U_{c,t-1}}\,\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}$$

Define $\widehat{U}_{c,t} \equiv U_{c,t}/(1-\chi_t)$*, with* $\chi_t \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}}\frac{\tau^y_t}{1-\tau^y_t}$ *as the complementarity-weighted labor wedge. Savings satisfy the generalized inverse Euler equation*

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right]$$

The commodity wedge is

$$\frac{\tau^q}{1+\tau^q} = \frac{\gamma_{c,t}-\gamma_{q,t}}{1+\epsilon_t-\gamma_{c,t}}\,\frac{\tau^y}{1-\tau^y}$$

Proof. See Appendices A.1, A.2, and A.3. □

Corollary 4.1 (dynamic Atkinson-Stiglitz).
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$*, the commodity wedge vanishes at every history,* $\tau^q_t = 0$*: the optimum places no wedge between the two goods, and differential commodity taxation adds nothing to the nonlinear income tax.*

Proof. Immediate from the commodity wedge of Result 4.1 at $\gamma_{c,t} = \gamma_{q,t}$. □

The labor wedge is Golosov et al. (2016)'s decomposition term for term, now written in the marginal utility of the numéraire. Three of the four pieces are unchanged: $A_t = 1 + \epsilon_t - \gamma_{c,t}$ scales the wedge by the responsiveness of labor supply, B_t is the hazard ratio that weighs the benefit of relaxing the incentive constraint at θ, which accrues to the mass $1 - F_t(\theta)$ of higher types, against its cost at the local mass $\theta f_t(\theta)$, and D_t carries the backward-looking persistence inherited from past distortions. Only C_t changes. C_t is the social value of the resources extracted from types above θ, each weighted by $1 - \lambda_{1,t}\bar{\alpha}_t U_{c,t}$, with $\lambda_{1,t}$ the multiplier on promise keeping that converts utils into resources (Appendix A.1). Its exponent measures how the numéraire's marginal utility evolves as the planner takes resources up the type distribution, and with two goods that evolution runs through both margins: the cross-curvature $\sigma_{cq,t}\dot{q}_t/q_t$ now sits beside the own-curvature $\sigma_t\dot{c}_t/c_t$. The reason is immediate: taking resources from a higher type cuts its consumption of both goods and, when $U_{cq} \neq 0$, the fall in the dirty good moves the marginal value of the numéraire in which the wedge is denominated. Under separability between the goods the cross term vanishes and C_t returns to its one-good form. The second good thus leaves the structure of the labor wedge intact, merely reweighting the income effect inside C_t.

The savings wedge is once more an inverse Euler equation, now in the incentive-adjusted marginal utility $\widehat{U}_{c,t} = U_{c,t}/(1 - \chi_t)$. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources; when the numéraire is complementary with labor, the same unit also lowers the information rent $-U_l\, l/\theta$ that truth-telling commands, an incentive gain worth $\chi_t/U_{c,t}$, so the net cost of delivery is $(1 - \chi_t)/U_{c,t}$. The planner equalizes the present value of this net cost across dates and states, exactly as the classical inverse Euler equation equalizes $1/U_c$ under separability. The adjustment χ_t is the complementarity-weighted labor wedge: it grows with the concurrent labor distortion, vanishes when the numéraire separates from labor, and reflects the numéraire's complementarity with labor, not the presence of the second good.

Hellwig (2021) characterizes the same margin with the adjustment placed in the probability measure instead. His generalized inverse Euler equation reweights the type density by the kernel $\exp[-\int U_{\theta c}/U_c]$ and scales the condition by a factor proportional to the current labor wedge. In the ability-only economy here, $U_{\theta c}/U_c = -\gamma_{c,t}/\theta$, so his reweighting kernel is, up to normalization, the integrating factor $\exp[-\int \gamma_{c,t}\, d\tilde{x}/\tilde{x}]$ that solves the costate equation of Appendix A.1. Resolving the costate against the labor wedge before stating the Euler equation converts his change of measure into the pointwise factor $1 - \chi_t$, so the condition holds under the original measure; the history dependence his scaling factor tracks with a persistence forecast rides here inside τ_t^y, through D_t. Both formulations collapse to the textbook rule once consumption and labor separate.

The commodity wedge is the labor wedge scaled by the ratio of the complementarity gap to the labor response. Its sign follows the Corlett-Hague logic: the planner taxes the good that is the weaker complement of labor. What is new relative to the static Mirrleesian commodity-tax problem is that this wedge is a dynamic object, history-dependent whenever preferences are nonseparable.

The dirty good has so far been only a second consumption good. Once its consumption pollutes, the planner acquires a corrective motive alongside the redistributive one, and the question is whether the two tangle. The next section shows they do not.

5 The optimal carbon wedge

Now let the dirty good pollute. Its consumption feeds the stock S_t, which enters utility directly, adding a corrective motive to the planner's redistributive one. This section states the separation of the two, then signs the deviations that arise when it fails. As before, the incentive constraints reduce to a recursive problem, following Kapička (2013). The state S_- collects the lagged aggregate emissions $(Q_{t-s}, ..., Q_{t-1})$ that still feed current and future stocks, and the continuation problem receives its update, with the current aggregate Q_t appended; I write both as single arguments to keep the notation light.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, S_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, S) \right] f_t(\theta|\theta_-) d\theta$$

$$\text{s.t.}$$

$$\dot{u}(\theta) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta|\theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta$$

$$u(\theta) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w(\theta)$$

$$S = S \left(\cdots, \int q(\theta) f(\theta) d\theta \right)$$

Solving the problem leaves the income-side wedges untouched. The externality adds a single corrective term to the first-order condition for the dirty good and changes nothing in the conditions for the numéraire, labor, and the costate (Appendix B.1), so the labor and savings wedges are exactly those of Result 4.1. Only the carbon wedge is new, and under weak separability it takes the cleanest possible form.

Proposition 5.1 (three-instrument separation).
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrective and redistributive margins separate. The

carbon wedge equals the social cost of carbon, the labor wedge keeps its Result 4.1 form, and the savings wedge keeps its inverse Euler form,

$$\tau_t^q = SCC_t, \qquad \frac{\tau_t^y}{1-\tau_t^y} = A_t B_t C_t + \beta R \, \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \, D_t, \qquad \frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \, \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right].$$

The carbon price carries no redistributive term; the income tax and the savings wedge do all the redistribution and insurance.

Proof. The first-order conditions for the numéraire, labor, and the costate are those of the no-pollution problem (Appendix B.1), so the labor and savings wedges are unchanged from Result 4.1. Result 5.1 gives $(\tau_t^q - \mathrm{SCC}_t)/(1+\tau_t^q) = (\gamma_{c,t} - \gamma_{q,t})[\,\cdot\,]$, which vanishes at $\gamma_{c,t} = \gamma_{q,t}$, leaving $\tau_t^q = \mathrm{SCC}_t$. □

This is the paper's central result. Redistributing through an optimal nonlinear income tax restores a split the Ramsey second best breaks: the carbon price corrects the externality and nothing else, while the income tax carries distribution. A dynamic Ramsey planner with the same fiscal needs, as in Barrage (2019) or Douenne et al. (2026), prices carbon below marginal damages, divided down by the marginal cost of public funds. Here the markdown is absent: it belonged to the linear instrument set the Ramsey analysis assumes, not to distortionary taxation itself, and it does not survive an optimal nonlinear income tax. The separation is a statement about the structure of the carbon wedge, not the level of the price: no screening term or Pareto weight enters its formula, but the social cost of carbon it equals is still an equilibrium object, with U_S/U_c evaluated at the redistributive optimum, so redistribution moves the level of the price even as it drops out of the formula.

The social cost of carbon in the Proposition is itself the textbook Pigouvian price when abatement does not interact with the labor distortion, either because the pollution stock is as complementary with labor as the clean good at every date, $\xi_{t+j} = \gamma_{c,t+j}$, or because labor is undistorted. It is then

$$\mathrm{SCC}_t = -\sum_{j=0}^{s} \frac{1}{R^j} \, \mathbb{E}_t \int \frac{U_{S,t+j}}{U_{c,t+j}} \, \Phi_j \, f_{t+j} \, d\theta,$$

the discounted expected stream of marginal damages valued in the numéraire, with $\Phi_j = \partial S_{t+j}/\partial Q_t$ the pass-through of current emissions to the future stock. Under weak separability and either of these conditions the carbon wedge is this Pigouvian price exactly, set with no reference to the fiscal system. The two conditions hold together when utility takes the form $U = V(h(c,q,S),l)$, the goods and the stock inside a common aggregate separable from labor: this is the dynamic image of the condition under which Jacobs and de Mooij (2015) recover the first-best Pigouvian rule under a nonlinear income tax (their Corollary 2). Otherwise SCC_t carries a fiscal-adjustment factor, given with the corrective wedge below.

Away from weak separability the carbon wedge deviates from the social cost of carbon, in a direction the complementarity gap signs.

Result 5.1 (Optimal corrective wedge)**.**
The optimal corrective wedge between the clean good and the polluting good is

$$\frac{\tau_t^q - SCC_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right]$$

where

$$SCC_t = -\sum_{j=0}^{s} \frac{1}{R^j} \, \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1-\tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1+\epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \, \Phi_j \, f_{t+j} \, d\theta$$

with $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ the effect of current aggregate emissions on the date-$(t+j)$ pollution stock ($\Phi_j = 0$ for $j > s$ by construction) and the expectation taken over future skill histories. The right-hand side is the no-pollution commodity wedge of Result 4.1 term for term, with A_t, B_t, C_t, D_t the labor-wedge terms. The externality changes only the left-hand side, replacing τ_t^q with the deviation $\tau_t^q - SCC_t$.

Proof. See Appendix B.1. □

The corrective wedge decomposes into two economically distinct components: a Mirrleesian screening component with the shape of the no-pollution wedge, and a corrective component built on the social cost of carbon SCC_t. The wedge now operates on the deviation $\tau_t^q - \mathrm{SCC}_t$ rather than on τ_t^q itself. The fiscal-adjustment factor on each damage term is what separates SCC_t from the Pigouvian benchmark written above. When $\xi_{t+j} > \gamma_{c,t+j}$, abatement relaxes the labor distortion and lends a fiscal bonus, so the SCC rises above the Pigouvian level; when $\xi_{t+j} < \gamma_{c,t+j}$, abatement tightens the distortion and the SCC falls below it. This is the "virtual subsidy" of Jacobs and de Mooij (2015), carried to a dynamic setting: their modified Samuelson rule scales each type's willingness to pay for environmental quality by one plus a term multiplying the labor wedge by $\partial \ln(u_E/u_c)/\partial \ln l$, and that derivative is exactly the gap $\xi - \gamma_c$ here. The planner over-abates, relative to pure Pigouvian pricing, when a cleaner environment complements work, and under-abates when it complements leisure.

6 The uniform carbon wedge

The commodity wedge of Sections 4 and 5 is history-dependent: it tracks the labor wedge node by node. Real commodity and carbon taxes are not. An excise or a carbon price is an anonymous linear price, a single figure per period that every buyer faces regardless of earnings history. This section re-solves the economy under exactly that restriction. The planner keeps the fully nonlinear, history-dependent income and savings instruments, but the consumer price of the dirty good must be uniform within each period. I ask how all three margins are set and what the restriction costs.

The restriction is a wedge constraint. A common consumer price $1+\bar{\tau}_t$ means the commodity wedge satisfies $U_q/U_c = 1+\bar{\tau}_t$ at every node, with the level $\bar{\tau}_t$ still chosen by the planner, so I append the restriction to the recursive problem as a pointwise constraint and price it with a multiplier. The multiplier is the section's organizing statistic: every condition below is its unconstrained counterpart plus a term proportional to it, the optimal uniform level sets its population average to zero, and Section 7 reuses the same device to freeze wedges at schedules that are neither uniform nor optimal. What survives of the one-good reading is a lemma: at a common price the agent's split of expenditure is a demand system, which is what keeps the first-order approach on its one-good footing and what the linear-commodity solver of Section 8 implements.

6.1 The restricted problem and its multiplier

The planner sets a uniform per-unit tax $\bar{\tau}_t$ on the dirty good. Its producer price is one, so its consumer price is $1+\bar{\tau}_t$, common to all agents at date t. The planner solves the problem of Section 4 (the externality waits until the end of the section) with the wedge constraint appended at every type,

$$\frac{U_q\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}\right)}{U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}\right)} = 1+\bar{\tau}_t \qquad \text{for all } \theta, \tag{1}$$

with multiplier $\eta^q(\theta)\, f_t(\theta|\theta_-)$ and with the level $\bar{\tau}_t$ itself a date-t choice. The realized commodity wedge is $1+\tau_t^q = U_q/U_c = 1+\bar{\tau}_t$, common to every agent: the uniform wedge is the uniform tax. The constraint involves neither u nor the continuation values, so the envelope condition, promise keeping, and the costate equation are untouched, and η^q measures the resource value, per agent of type θ, of a marginal rise in that type's consumer price.

Although the planner assigns both goods directly, the constraint makes the assignment read as demand: any bundle satisfying (1) is the split an agent facing the common price would choose of the expenditure $x \equiv c + (1+\bar{\tau}_t)q$, with Marshallian demands $c^d(x, l, \bar{\tau})$ and $q^d(x, l, \bar{\tau})$ pinned down by the tangency $U_q = (1+\bar{\tau})U_c$ and the budget. Three statistics of this demand system organize the results,

$$m_q \equiv (1+\bar{\tau})\,\partial_x q^d, \qquad s_{qq} \equiv \left.\frac{\partial q^d}{\partial(1+\bar{\tau})}\right|_u = \frac{U_c}{\mathcal{D}} < 0, \qquad \mathcal{I}_t \equiv \frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{(1+\tau_t^q)\,q_t}.$$

m_q is the marginal budget share of the dirty good, the fraction of a marginal dollar of expenditure spent on it. s_{qq} is its compensated own-price response, negative by the strict quasi-concavity $\mathcal{D} \equiv U_{qq} - 2(1+\bar{\tau})U_{cq} + (1+\bar{\tau})^2 U_{cc} < 0$, which I assume throughout. And the income-effect gap $\mathcal{I}_t$, evaluated here at the common price $1+\tau_t^q = 1+\bar{\tau}_t$, is positive exactly when the dirty good is normal, since $\partial_x q^d = (1+\bar{\tau})\,U_c \mathcal{I}_t/|\mathcal{D}|$ (Appendix C.3).

Lemma 6.1 (implementation and the first-order approach).
An allocation satisfies (1) *if and only if it assigns each type the private split of some pair* (x, y) *of expenditure and output at the common price, with indirect utility*

$$\tilde{U}(x, l; \bar{\tau}) \equiv \max_{c,q\geq 0} U(c, q, l) \qquad s.t. \qquad c + (1+\bar{\tau})q \leq x;$$

the split is unique under $\mathcal{D} < 0$*, and at it* $\tilde{U}_x = U_c$ *and* $\tilde{U}_l = U_l$*. Incentive compatibility in the restricted economy is therefore that of a one-good economy in* (x, y) *with period utility* $\tilde{U}(x, y/\theta; \bar{\tau}_t)$*, and the envelope condition is*

$$\dot{u}(\theta) = -\frac{\tilde{U}_l\, l}{\theta} + \beta\, w_2(\theta) = -\frac{U_l\, l}{\theta} + \beta\, w_2(\theta).$$

Proof. See Appendix C.2. □

The lemma's work is foundational rather than computational. A deviator who misreports receives the reported type's expenditure-output pair and splits it at the same common price, so the deviation space is that of a one-good economy in (x, y), and the relaxed-problem apparatus of Section 3 applies on the one-good footing it was built for, without the multi-good sufficiency caveat. The same reduction is the object the quantitative section solves: the linear-commodity solver of Section 8 assigns expenditure and lets the demand system split it. The derivations, though, run through the multiplier. The ratio $\psi\, U_{c,t}/(\theta f_t)$ is the normalized shadow price of the local incentive constraint, with ψ the costate on the envelope condition as in Appendix A.1; in this notation the unconstrained commodity wedge of Appendix A.3 is $\tau^q/(1+\tau^q) = (\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f}$.

Lemma 6.2 (the multiplier).
At every history,

$$\eta_t^q = (1+\bar{\tau}_t)\,|s_{qq,t}|\left[(\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t} - \frac{\bar{\tau}_t}{1+\bar{\tau}_t}\right].$$

Proof. See Appendix C.4. □

The bracket is the gap between the commodity wedge the planner would set at the node if the instrument were free, $(\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f}$, and the wedge the uniform tax imposes; the compensated substitution response converts the gap into resources. The multiplier is positive where the uniform tax falls short of the node's Corlett-Hague level, negative where it overshoots, and everything the restriction does to the remaining margins runs through it. Substituting it into the first-order conditions assembles the primitive statistics into composites, the curvatures a reader of the one-good reduction would compute from the indirect utility $\tilde{U}$.

Lemma 6.3 (composite statistics).
Define

$$\tilde{\gamma} \equiv (1 - m_q)\,\gamma_c + m_q\,\gamma_q, \qquad \tilde{\epsilon} \equiv \epsilon - \frac{(1+\bar{\tau})^2\,U_c^2\,(\gamma_c - \gamma_q)^2}{l\,|U_l|\,|\mathcal{D}|} \;\leq\; \epsilon, \qquad \tilde{A} \equiv 1 + \tilde{\epsilon} - \tilde{\gamma}.$$

Then $\tilde{\gamma} = \tilde{U}_{xl}\,l/\tilde{U}_x$ *and* $\tilde{\epsilon} = \tilde{U}_{ll}\,l/\tilde{U}_l$ *for the indirect utility of Lemma 6.1. Under weak separability,* $\gamma_c = \gamma_q$*, so* $\tilde{\gamma} = \gamma_c = \gamma_q$*,* $\tilde{\epsilon} = \epsilon$*, and the restricted problem is that of Section 4 exactly.*

Proof. See Appendix C.7. □

The composite complementarity is the marginal-spending-weighted average of the two goods' complementarities: a marginal dollar of expenditure is split $(1 - m_q, m_q)$ across the goods, so the marginal utility it buys co-moves with labor accordingly. The inequality on $\tilde{\epsilon}$ is a Le Chatelier property. The agent's freedom to re-optimize the split cushions labor shocks, so the effective disutility of labor is less convex, and delivering expenditure is a weakly better hedge than delivering a rigid bundle. Both corrections are proportional to the same complementarity gap $\gamma_c - \gamma_q$ that drives the unconstrained wedge of Section 4: everything the restriction does vanishes when the two goods are equally complementary with labor.

6.2 The three margins

The three margins follow from the first-order conditions, each carrying a fiscal-interaction term, proportional to the multiplier, that the unconstrained problem lacked.

Result 6.1 (labor wedge under a uniform commodity tax).
At every history,

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t\,\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^q\,(1+\bar{\tau}_t)\left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1-\tau_t^y)\,y_t}\right],$$

equivalently, in composite statistics,

$$\frac{\tau_t^y}{1-\tau_t^y} + \frac{\bar{\tau}_t}{1-\tau_t^y}\,\frac{dq_t^d}{dy}\bigg|_u = \tilde{A}_t\,\frac{\psi U_{c,t}}{\theta f_t}, \qquad \frac{dq^d}{dy}\bigg|_u \equiv (1-\tau^y)\,\partial_x q^d + \frac{\partial_l q^d}{\theta},$$

where $dq^d/dy|_u$ is the utility-compensated response of dirty-good demand to earnings: the change in q^d when output rises by one unit and expenditure rises by $(1-\tau^y)$ units so as to hold utility fixed.

Proof. See Appendices C.5 and C.7. □

The restriction sets a comprehensive wedge, not the income-tax wedge alone. When an agent earns a marginal unit of output on a compensated path, the planner collects τ_t^y through the income margin and $\bar{\tau}_t\, dq^d/dy|_u$ through the commodity margin; their sum $\Omega_t \equiv \tau_t^y/(1-\tau_t^y) + \bar{\tau}_t/(1-\tau_t^y) \cdot dq^d/dy|_u$ is what the screening condition pins down. The correction has two parts of opposite sign. The expenditure part $(1-\tau^y)\,\partial_x q^d$ is positive for a normal good: compensated earnings raise spending, and the fraction m_q of the marginal dollar returns as commodity revenue. The labor part $\partial_l q^d/\theta$ is the statistic of Christiansen (1984), negative when the dirty good is a leisure complement, $\gamma_q < \gamma_c$: working more at fixed expenditure shifts spending away from the good. When the leisure-complement part dominates, the measured income-tax wedge exceeds the naive $\tilde{A}_t\, \psi U_c/(\theta f)$, and the economics is that discouraging labor is fiscally cheaper: the agent who works less spends more on the taxed good, so part of the forgone output returns as commodity revenue. When the expenditure part dominates, the correction runs the other way, and the income-tax wedge sits below $\tilde{A}_t\, \psi U_c/(\theta f)$.

Result 6.2 (optimal uniform commodity tax)**.**
The optimal level of the uniform tax sets the population average of the multiplier to zero at every date, $\mathbb{E}_0[\eta_t^q] = 0$; equivalently

$$\bar{\tau}_t\ \mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\, l_t\, \partial_l q_t^d\right] \qquad \Longleftrightarrow \qquad \frac{\tau_t^q}{1+\tau_t^q} = \frac{\mathbb{E}_0\left[|s_{qq,t}|\,(\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]},$$

where $\mathbb{E}_0$ is the unconditional population expectation over date-t histories θ^t. The uniform ad-valorem wedge is the substitution-weighted average of the unconstrained node-specific wedge $(\gamma_c-\gamma_q)\,\psi U_c/(\theta f)$ of Section 4.

Proof. See Appendix C.9. □

The single instrument averages the schedule the planner would set node by node if it could, weighting each history by the compensated substitutability $|s_{qq}|$ at that node. Nodes where the consumer can easily substitute away from the dirty good carry more weight, exactly as in a many-person Ramsey rule. Two limits are immediate.

Corollary 6.1 (when uniformity is costless)**.**
If preferences are weakly separable, $\gamma_c = \gamma_q$ pointwise, then $\bar{\tau}_t = 0$ for every t, the multiplier vanishes at every node, and the restricted optimum coincides with the unconstrained optimum.

This is the dynamic Atkinson-Stiglitz benchmark of Corollary 4.1. More generally the restriction is costless at date t if and only if the unconstrained wedge $(\gamma_c - \gamma_q)\,\psi U_c/(\theta f)$ is constant across the date-t cross-section.

Corollary 6.2 (retirement).
Suppose agents retire at a date $T_R \leq T$: from T_R on there is no labor to report, so $\psi \equiv 0$, hence $\psi U_c/(\theta f) \equiv 0$ and $\bar{\tau}_t = 0$. The screening component of the uniform tax dies with the labor margin; with the externality below, the dirty-good price converges to pure Pigouvian pricing for retirees.

Result 6.3 (savings wedge under a uniform commodity tax).
Define the incentive-plus-fiscal adjustment

$$\chi_t^F \equiv \underbrace{\tilde{\gamma}_t \frac{\psi U_{c,t}}{\theta f_t}}_{incentive} + \underbrace{\varpi_t\, m_{q,t}}_{fiscal}, \qquad \varpi_t \equiv \frac{\bar{\tau}_t}{1+\bar{\tau}_t},$$

equal to the primitive form $\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q\,(1+\bar{\tau}_t)\,\mathcal{I}_t$. The generalized inverse Euler equation holds in the adjusted marginal utility $U_{c,t}/(1-\chi_t^F)$,

$$\frac{1-\chi_t^F}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^F}{U_{c,t+1}}\right],$$

and to second order $\tau_t^s \approx \mathrm{Var}_t(\ln U_{c,t+1}) + (\chi_t^F - \mathbb{E}_t\,\chi_{t+1}^F) + \mathrm{Cov}_t(\chi_{t+1}^F, \ln U_{c,t+1})$.

Proof. See Appendix C.6. □

The savings adjustment now has two parts. The incentive part $\tilde{\gamma}_t\,\psi U_c/(\theta f)$ is the Section 4 adjustment on the composite margin, present whenever expenditure is complementary with labor. The fiscal part $\varpi_t m_{q,t}$ is new. Delivering a marginal unit of expenditure returns the fraction $\varpi_t m_{q,t}$ to the planner as commodity revenue, so its net resource cost is $1 - \varpi_t m_{q,t}$, and the planner equalizes the present value of this net, incentive-adjusted cost across dates and states. Through the second-order expression the savings wedge picks up the term $\varpi_t m_{q,t} - \mathbb{E}_t[\varpi_{t+1} m_{q,t+1}]$, the expected change in the commodity-revenue content of spending. With a positive tax on a leisure complement, agents spend a larger share on the taxed good precisely when leisure is high late in the working life, so this term tends to be negative: deferring expenditure to high-leisure working years generates future commodity revenue, and the planner credits saving for it. The credit stops at retirement, where Corollary 6.2 sets the screening tax, and with it ϖ, to zero.

The comprehensive wedge Ω_t inherits the full dynamic structure of the Section 4 labor wedge.

Result 6.4 (comprehensive-wedge decomposition).
At every history,

$$\Omega_t = \tilde{A}_t(\theta)\,\tilde{B}_t(\theta)\,\tilde{C}_t(\theta) + \beta R\,\Omega_{t-1}\,\tilde{D}_t(\theta),$$

where $\tilde{B}_t = B_t$ is the hazard ratio of Result 4.1 and $\tilde{C}_t, \tilde{D}_t$ are its C_t and D_t with the integrating kernel built on the composite $\tilde{\gamma}$ and, in $\tilde{C}_t$, the redistributive weight carrying the revenue term $1 - \bar{\tau}_t\,\partial_x q^d - \lambda_{1,t}\bar{\alpha}_t U_{c,t}$.

Proof. See Appendix C.8. □

All of the qualitative structure survives the restriction: the hazard-driven shape through $\tilde{B}_t$, the persistence channel through $\tilde{D}_t$. What it now governs is the comprehensive margin Ω_t. Result 6.1 then dictates, given $\bar{\tau}_t$, how Ω_t splits between the income-tax wedge and the commodity-revenue term.

6.3 What uniformity costs, and where it binds

With the margins set, what does the restriction cost? Because a single rate matches the personalized schedule only on average, the loss is second-order: it grows with the dispersion of the wedge the planner would otherwise set node by node, weighted at each history by how easily the consumer can substitute away from the dirty good.

Result 6.5 (cost of uniformity).
Let $\varpi_t^*(\theta^t) \equiv (\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}$ denote the unconstrained node-specific ad-valorem wedge, and let $\mathrm{Var}_t^{|s|}$ denote the $|s_{qq,t}|$-weighted cross-sectional variance over date-t histories. To second order in the dispersion of ϖ_t^*, the date-t resource-equivalent loss from imposing a common tax rather than the node-specific schedule is

$$\mathrm{Loss}_t \approx \frac{1}{2}\,(1+\bar{\tau}_t)^3\;\mathbb{E}_0[|s_{qq,t}|]\;\mathrm{Var}_t^{|s|}(\varpi_t^*)\,,$$

minimized at the uniform rate of Result 6.2.

Proof. See Appendix C.10. □

The cost of not personalizing the commodity tax is the substitution-weighted cross-sectional variance of the unconstrained wedge. It vanishes under weak separability, where $\varpi^* \equiv 0$, and it is small whenever the complementarity gap is small or the wedge $\varpi^* = (\gamma_c - \gamma_q)\,\psi U_c/(\theta f)$ is flat across the cross-section. This is the proportionality result of Section 4 cashed out: because the unconstrained wedge tracks the labor wedge, its dispersion is first-order in the dispersion of τ^y, and what that tracking is worth is exactly what uniformity forgoes. Every object in the formula is a demand-side or tax-side statistic, so it is a sufficient-statistics answer to the cost-of-mispricing question, and the near-sufficiency of a flat carbon price that the quantitative section measures is this variance being small.

Now reintroduce the externality of Section 5. Because the stock is aggregate and deterministic, it is invariant to reports and acts as a dated preference shifter; the dirty good's first-order condition gains the shadow emission price $\nu_t = \mathrm{SCC}_t$ of Appendix B.1, and every condition of this section holds with the deviation $\bar{\tau}_t/(1+\bar{\tau}_t)$ replaced by $(\tau_t^q - \mathrm{SCC}_t)/(1+\tau_t^q)$, where $1+\tau_t^q$ is the common consumer price. The uniform per-unit tax net of marginal damages is the screening component $\bar{\tau}_t^{\text{screen}} \equiv \tau_t^q - \mathrm{SCC}_t$, the revenue wedge net of the part that pays for damages.

Result 6.6 (additive split of the uniform price)**.**
At an interior optimum of the restricted economy with the externality, all of the preceding results hold with $\bar{\tau}_t = \tau_t^q - \mathrm{SCC}_t$ the screening tax and every statistic evaluated at the externality allocation. The optimal uniform consumer price splits additively,

$$1+\tau_t^q = 1 + \mathrm{SCC}_t + \bar{\tau}_t^{\text{screen}}, \qquad \frac{\bar{\tau}_t^{\text{screen}}}{1+\tau_t^q} = \frac{\mathbb{E}_0\left[|s_{qq,t}|\left(\gamma_{c,t}-\gamma_{q,t}\right)\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0\left[|s_{qq,t}|\right]},$$

and the social cost of carbon keeps its discounted-damage form, its damage weight adding to the Section 5 fiscal factor a term proportional to the multiplier η_t^q (Appendix C.11).

Proof. See Appendix C.11. □

The uniformity restriction never binds the Pigouvian component. The corrective part of the optimal dirty-good price was uniform already at the unconstrained optimum, since a ton of emissions carries the same social cost whoever emits it and the emissions margin cannot screen. What uniformity constrains is only the Corlett-Hague screening part, forced from a history-dependent schedule down to the substitution-weighted scalar of Result 6.2. For carbon policy the reading is sharp: a legislated uniform carbon price sacrifices nothing on the corrective margin, whatever is lost is lost on screening, and Result 6.5 measures that loss. This is the separation of Section 5 restated for the instrument governments actually use.

One caveat is proper to a restricted-instrument setting. The full-optimum production-efficiency logic of Diamond and Mirrlees (1971) no longer applies automatically once an instrument is constrained, and Naito (1999) shows that a constrained planner can then gain from distorting production. Under the linear technology maintained throughout, the skill types are perfect substitutes and relative wages cannot move, so production efficiency holds automatically and the results of this section are exact. If clean and dirty production employed different skill mixes, the constrained optimum would generically feature prices that differ across sectors and incidence terms from general equilibrium (Stiglitz 1982; Sachs et al. 2020).

7 Third-best wedges

The separation of Proposition 5.1 is a property of the jointly optimal tax system. The carbon price sheds its redistributive role because the labor wedge, set optimally node by node, absorbs it. Real income taxes are nothing like that: they are legislated, coarse, and revised on political time, so the labor wedge they imply follows a schedule the planner does not choose. This section freezes one wedge at a time at such a schedule and re-optimizes the remaining margins against it: the labor wedge, then the savings wedge, then the commodity wedge itself. I call the resulting problem the third best, in the sense of van der Ploeg et al. (2025): the second best restricts information, the third best also restricts an instrument.

The device is Section 6's: a wedge constraint appended to the recursive problem, with a multiplier pricing the gap between the frozen wedge and the one the planner would set. There the frozen level was chosen optimally, so the multiplier averaged to zero; here the schedules are arbitrary, and the multiplier's profile is the object of interest. For the labor case, two results follow. The savings wedge keeps its inverse-Euler form in an adjusted marginal utility, but the adjustment now prices an income effect: delivering the numéraire depresses labor supply, and the frozen schedule can no longer absorb the shift. And the carbon price loses its anchor. Even under weak separability it deviates from the social cost of carbon wherever the schedule misses the wedge the planner would set, falling below the social cost of carbon where the schedule overtaxes labor and rising above it where the schedule undertaxes.

Freezing the savings wedge instead restricts a ratio of marginal utilities across dates, and its error can only tilt the carbon price: the deviations from the social cost of carbon must take both signs over the life cycle. Freezing the commodity wedge reverses the flow of contamination: a carbon price legislated below the social cost of carbon raises the optimal labor wedge, by the marginal budget share of the dirty good times the uncollected carbon price, and moves the savings wedge with the drift of that charge. The contrast with Section 6 is sharp throughout: restricting the carbon instrument to a uniform price left the Pigouvian component intact, and freezing any wedge at its optimum is equally harmless. What the separation cannot survive is error, in any of the three instruments. A closing subsection turns the same machinery into a bridge to the Ramsey literature: on every margin, the optimal linear instrument is the level at which the constraint multiplier averages to zero.

7.1 A constrained labor wedge

The planner solves the problem of Section 5 with the labor wedge held to an exogenous schedule $\bar{\tau}_t^y(\theta) < 1$, a function of the date and the current type,

$$V_t(\hat{w}, \hat{w}_2, \theta_-, S_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, S)\right] f_t(\theta|\theta_-) d\theta$$

subject to the constraints of Section 5 and, at every type, the wedge constraint

$$\Gamma(\theta) \equiv -\frac{U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{\theta\, U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} - (1 - \bar{\tau}_t^y(\theta)) = 0. \tag{2}$$

The leading case for the schedule is the wedge implied by an actual tax-and-transfer system, read into type space through the equilibrium mapping from income to skill, but nothing below restricts its shape. The constraint has a decentralized reading: node by node, (2) is the labor-supply first-order condition of an agent free to choose hours at the marginal net-of-tax wage $(1 - \bar{\tau}_t^y)\theta$. The planner still assigns both goods and the intertemporal margin; the labor margin now clears on the agent's side at the frozen rate. The constraint changes no part of the incentive apparatus, since it involves neither u nor the continuation values, so the envelope condition, promise keeping, and the costate equation are those of Section 5. What it changes is feasibility, and the relaxed-problem caveat of Section 3 applies with extra force: a schedule chosen by politics has no reason to respect the monotonicity that global incentive compatibility asks of the allocation, so the verification step matters more here, not less.

Let $\eta(\theta)\, f_t(\theta|\theta_-)$ be the multiplier on (2), so that η is the resource value, per agent of type θ, of a marginal increase in the frozen net-of-tax rate. Two statistics organize the results, alongside the normalized shadow price $\psi\, U_{c,t}/(\theta f_t)$ of the local incentive constraint of Section 6, all evaluated at the constrained allocation, with $y = \theta l$:

$$\Gamma_{c,t} \equiv (1 - \bar{\tau}_t^y)\, \frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{y_t}, \qquad \Delta_t \equiv (1 - \bar{\tau}_t^y)\, \frac{\sigma_t}{c_t} + \frac{\epsilon_t - 2\gamma_{c,t}}{y_t}.$$

$\Gamma_{c,t} = \partial\Gamma/\partial c$ is the response of the implied net-of-tax rate to the numéraire; it is positive exactly when leisure is a normal good, so that delivering consumption depresses labor supply (Appendix D). Δ_t is the same response along a utility-compensated increase in consumption and labor; it is proportional to the second-order condition of the agent's hours choice at the frozen rate, so $\Delta_t > 0$ wherever the schedule decentralizes an interior labor choice, which I assume throughout (Appendix D.5).

Lemma 7.1 (the price of the frozen schedule)**.**
At every history,

$$\eta_t\, \Delta_t = \frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} - A_t\, \frac{\psi U_{c,t}}{\theta f_t},$$

with $A_t = 1 + \epsilon_t - \gamma_{c,t}$. The multiplier is proportional to the gap between the frozen wedge and the virtual wedge $A_t \frac{\psi U_c}{\theta f}$, the labor wedge the third-best costate would decentralize if the labor margin were free: $\eta_t > 0$ where the schedule overtaxes labor, $\eta_t < 0$ where it undertaxes, and $\eta_t = 0$ where it crosses the virtual optimum.

Proof. See Appendix D. □

The lemma is the section's engine. In the unconstrained problem, combining the first-order conditions for the numéraire and labor delivers the labor wedge, $\tau_t^y/(1-\tau_t^y) = A_t \frac{\psi U_c}{\theta f}$. Here the same combination cannot determine the wedge, which is frozen; it determines instead the shadow price of freezing it. The virtual wedge is not the Section 4 wedge: ψ solves the same costate equation with an integrating factor and a source term modified by the constraint (Appendix D.3), so $\psi U_c/(\theta f)$ carries the hazard shape and the persistence of the ABCD decomposition, evaluated in the third best. Setting $\eta_t \equiv 0$, which requires the schedule to coincide with the virtual wedge at every node, returns every formula of Section 5.

Result 7.1 (third-best savings wedge)**.**
Define $\chi_t^\eta \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \Gamma_{c,t}$ and the adjusted marginal utility $\widehat{U}_{c,t}^\eta \equiv U_{c,t}/(1-\chi_t^\eta)$. The generalized inverse Euler equation holds in $\widehat{U}_c^\eta$,

$$\frac{1}{\widehat{U}_{c,t}^\eta} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}^\eta}\right],$$

without uncertainty the savings wedge is $1-\tau_t^s = (1-\chi_t^\eta)/(1-\chi_{t+1}^\eta)$, and to second order $\tau_t^s \approx \mathrm{Var}_t(\ln U_{c,t+1}) + (\chi_t^\eta - \mathbb{E}_t\,\chi_{t+1}^\eta) + \mathrm{Cov}_t(\chi_{t+1}^\eta, \ln U_{c,t+1})$.

Proof. See Appendix D.1. □

The delivery-cost logic of Section 4 extends by one term. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources and, as before, earns the incentive credit $\gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t}$ when the numéraire is complementary with labor. The new term prices the income effect. The same unit of consumption moves the implied net-of-tax rate by $\Gamma_{c,t}$, and with the rate frozen, labor supply must fall to restore (2); the planner values that induced fall at η_t. Where the schedule overtaxes labor ($\eta_t > 0$), consumption delivery drags labor down at a node where labor is already too low, so its net cost rises and χ_t^η falls. The savings wedge then follows the drift of the drag. The deterministic form says the planner taxes savings where the schedule's overtaxation worsens with age and subsidizes it where the overtaxation eases: moving resources toward dates where consumption depresses an already overtaxed labor margin is discouraged. Under separability of the numéraire from labor, $\gamma_{c,t} = 0$, the unconstrained adjustment vanishes and the entire savings distortion is the schedule's error, $\chi_t^\eta = -\eta_t(1-\bar{\tau}_t^y)\sigma_t/c_t$.

Result 7.2 (third-best carbon wedge).
Let $\mathcal{I}_t$ be the income-effect gap of Section 6, positive exactly when the dirty good is normal, evaluated here at the endogenous price $1+\tau_t^q$. At every history,

$$\frac{\tau_t^q - SCC_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t})\left(\frac{\psi U_{c,t}}{\theta f_t} + \frac{\eta_t}{y_t}\right) - \eta_t\,(1-\bar{\tau}_t^y)\,\mathcal{I}_t,$$

where the social cost of carbon keeps its discounted-damage form with a third-best fiscal factor,

$$SCC_t = -\sum_{j=0}^{s}\frac{1}{R^j}\,\mathbb{E}_t\int\left[1+(\xi_{t+j}-\gamma_{c,t+j})\left(\frac{\psi U_{c,t+j}}{\theta f_{t+j}}+\frac{\eta_{t+j}}{y_{t+j}}\right)+\eta_{t+j}\left(1-\bar{\tau}_{t+j}^y\right)\frac{\sigma_{t+j}+\xi_{c,t+j}}{c_{t+j}}\right]\frac{U_{S,t+j}}{U_{c,t+j}}\,\Phi_j\,f_{t+j}\,d\theta,$$

with $\xi_{c,t} \equiv U_{Sc,t}\,c_t/U_{S,t}$ the complementarity of the pollutant with the numéraire.

Proof. See Appendices D.2 and D.4. □

Corollary 7.1 (the separation fails in the third best).
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau_t^q - SCC_t}{1+\tau_t^q} = -\,\eta_t\,(1-\bar{\tau}_t^y)\,\mathcal{I}_t,$$

so when the dirty good is normal, $\operatorname{sign}(\tau_t^q - SCC_t) = -\operatorname{sign}(\eta_t)$: the optimum prices carbon below the social cost of carbon where the frozen schedule overtaxes labor, and above it where the schedule undertaxes. The carbon wedge equals the social cost of carbon at every history only if the schedule coincides with the virtual wedge, $\eta_t \equiv 0$.

The mechanism is the income effect the second best never needed. In Section 5 the commodity wedge was a compensated object: with the labor wedge free, only the complementarity gap $\gamma_c - \gamma_q$, a substitution statistic, could move it, which is Christiansen (1984)'s finding that under an optimal income tax the commodity tax turns on compensated cross-effects alone. Freezing the income tax lets uncompensated effects back in. Delivering the numéraire depresses labor supply through its income effect; delivering the dirty good does so only through the cross-curvature σ_{cq}; the gap $\mathcal{I}_t$ between the two is positive when the dirty good is normal. Where the schedule overtaxes labor, deliveries that depress labor are penalized at the shadow price η_t, the planner tilts the assigned bundle toward the good that carries less of the drag, and the relative price of the dirty good falls below its corrective level. Where the schedule undertaxes, the tilt reverses. The carbon price is bent to patch the labor distortion the frozen schedule leaves behind. All three constraint statistics say the same thing in different margins: $\Gamma_{c,t}$, $\mathcal{I}_t$, and $(\sigma_t + \xi_{c,t})/c_t$ are the log-sensitivities to consumption of the implied net-of-tax rate, of the relative price of the dirty good, and of the marginal damage valuation U_S/U_c, the three prices the frozen margin drags on.

The corollary sharpens the reading of Kaplow (2012). His argument frees Pigouvian pricing from any optimality requirement on the income tax: whatever the schedule, an environmental reform can be packaged with a type-by-type income-tax adjustment that keeps the distribution whole, so moving to marginal-damage pricing is a Pareto improvement. That package is exactly what the third best forbids. With the labor wedge frozen node by node, the compensating adjustment is unavailable, and the deviation $-\eta_t(1-\bar{\tau}_t^y)\mathcal{I}_t$ prices the missing degree of freedom. The same rigidity reaches the social cost of carbon itself: its fiscal factor now carries the schedule's error, and even the $U = V(h(c,q,S),l)$ benchmark, which made the second-best SCC purely Pigouvian in Section 5, no longer does so unless $\eta_t = 0$. In the single-cohort reading of Section 8, where marginal damage is a constant resource cost, $\mathrm{SCC}_t = e$ exactly and the third-best deviations act on $\tau_t^q - e$ undiluted.

Two limits mirror earlier results. At nodes where the schedule crosses the virtual wedge, $\eta_t = 0$ and the carbon wedge reverts to SCC_t plus the Corlett-Hague term $(\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f}$. And in retirement there is no labor to report or to distort: the wedge constraint is vacuous, $\psi \equiv 0$ and $\eta \equiv 0$, the savings wedge reverts to the standard inverse Euler form, and retirees face pure Pigouvian carbon pricing, as in Corollary 6.2. The formulas are otherwise measurement-ready: given the schedule an actual tax-and-transfer system implies, η_t is computable from solved allocations of the kind Section 8 produces, and with it the sign and size of the carbon correction node by node.

The uniform-tax exercise of Section 6 also has a labor counterpart: freeze the schedule flat and choose its level.

Result 7.3 (the optimal flat labor tax).
Let the schedule be flat, $\bar{\tau}_t^y(\theta) = \bar{\tau}_t^y$, with its level chosen optimally at every working date. The optimal level sets the population average of the multiplier to zero, $\mathbb{E}_0[\eta_t] = 0$; equivalently,

$$\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} = \frac{\mathbb{E}_0\left[s_{yy,t}\, A_t\, \frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[s_{yy,t}]}, \qquad s_{yy,t} \equiv \left.\frac{\partial y}{\partial\,(1-\bar{\tau}_t^y)}\right|_u = \frac{1}{(1-\bar{\tau}_t^y)\,\Delta_t},$$

with $s_{yy,t}$ the compensated response of earnings to the net-of-tax rate.

Proof. See Appendix D.5. □

The optimal flat rate is the compensated-earnings-response-weighted average of the virtual Mirrleesian wedges, the exact mirror of Result 6.2 with s_{yy} in place of $|s_{qq}|$: a single rate averages the schedule the planner would set node by node, weighting each node by how elastically its earnings respond. This is the optimal linear income tax of the Ramsey tradition, rendered in mechanism-design statistics, with one caveat: the constraint fixes the marginal rate while the mechanism still assigns consumption, so the implied instrument is a flat rate with history-dependent intercepts rather than a single linear budget line. Section 7.4 collects these zero-average conditions and their Ramsey readings.

7.2 A constrained savings wedge

Now freeze the intertemporal margin instead. The labor and commodity wedges are free, and the savings wedge must follow an exogenous schedule $\bar{\tau}^s_t(\theta) < 1$, node by node,

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}^s_t(\theta_t))\, \beta R\ \mathbb{E}_t\big[U_{c,t+1}(\theta^{t+1})\big]\,. \tag{3}$$

This is the agent's Euler equation at the frozen net return $(1-\bar{\tau}^s_t)R$: the intertemporal margin now clears on the agent's side, as the labor margin did in Section 7.1. The benchmark against which the schedule errs is not zero, since the unconstrained optimum itself carries a positive savings wedge through the inverse Euler equation (Golosov et al. 2003). A real capital income tax fixes an instrument rather than a wedge, so reading it into $\bar{\tau}^s_t(\theta)$ again uses the equilibrium mapping, as with the labor schedule.

Unlike the labor constraint, (3) is intertemporal: it ties the date-t allocation to expected marginal utility one period ahead, so it cannot enter the period Hamiltonian pointwise. It fits the recursive problem through one more state, a marginal-utility promise. The date-t problem passes each successor the required expectation $M(\theta) \equiv U_{c,t}(\theta)/((1 - \bar{\tau}^s_t(\theta))\, \beta R)$ and inherits its own requirement $\hat{M}$,

$$V_t(\hat{w}, \hat{w}_2, \hat{M}, \theta_-, S_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), M(\theta), \theta, S)\right] f_t(\theta|\theta_-)\, d\theta$$

subject to the constraints of Section 5 and the inherited requirement

$$\hat{M} = \int_0^\infty U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right) f_t(\theta|\theta_-)\, d\theta, \tag{4}$$

with multiplier ϑ_t, so that $\partial V_t/\partial \hat{M} = -\vartheta_t$; at $t = 0$ there is no inherited requirement and $\vartheta_0 = 0$, and no requirement is passed from the final period. Raising the marginal utility of the numéraire at a node now has a price of its own: it serves the inherited requirement, worth ϑ_t per util, and it raises the requirement passed forward, costing $\vartheta_{t+1}(\theta^t)/((1 - \bar{\tau}^s_t)\, \beta R^2)$. Their net, made dimensionless, is the subsection's multiplier,

$$\eta^s_t(\theta^t) \equiv \left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}^s_t)\, \beta R^2}\right] U_{c,t}(\theta^t).$$

Every first-order condition gains one term, η_t^s times the log-derivative of $U_{c,t}$ in the corresponding direction (Appendix E.1). The counterpart of Lemma 7.1 is now a law of motion rather than a static condition.

Lemma 7.2 (the multiplier's law of motion).
Define $\chi_t^s \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s\,\sigma_t/c_t$. The planner's intertemporal condition keeps the inverse-Euler form in $U_{c,t}/(1-\chi_t^s)$,

$$\frac{1-\chi_t^s}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\right],$$

and combining it with (3) pins the multiplier's drift. When the date-$(t+1)$ draw is degenerate, as in retirement,

$$\eta_t^s\,\frac{\sigma_t}{c_t} - (1-\bar{\tau}_t^s)\,\eta_{t+1}^s\,\frac{\sigma_{t+1}}{c_{t+1}} = \left(1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}\right)(\bar{\tau}_t^s - \tau_t^{s,*}),$$

where the virtual savings wedge is defined by $1-\tau_t^{s,*} \equiv (1-\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t})/(1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}})$, with initial condition $\vartheta_0 = 0$ and terminal value $\eta_T^s = \vartheta_T\,U_{c,T}$.

Proof. See Appendix E.2. □

The frozen wedge does not break the planner's inverse-Euler logic; it survives in the adjusted marginal utility, and the adjustment absorbs the schedule's error. What changes relative to Section 7.1 is what the error pins down. A frozen labor wedge is a static restriction, so its gap sets the multiplier's level node by node. A frozen savings wedge restricts a ratio of marginal utilities across dates, so its gap injects drift into η_t^s, and the level is tied down at the two ends of life, $\vartheta_0 = 0$ at birth and $\eta_T^s = \vartheta_T U_{c,T}$ at the close. Setting $\bar{\tau}_t^s = \tau_t^{s,*}$ at every date solves the law of motion with $\eta^s \equiv 0$ and returns every formula of Section 5.

Result 7.4 (wedges under a frozen savings wedge).
At every history,

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t\,\frac{\psi U_{c,t}}{\theta f_t} - \eta_t^s\left[\frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{(1-\tau_t^y)\,y_t}\right], \qquad \frac{\tau_t^q - SCC_t}{1+\tau_t^q} = (\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s\,\mathcal{I}_t,$$

with $\mathcal{I}_t$ the income-effect gap of Section 6. The social cost of carbon keeps its discounted-damage form, with the damage weight

$$1 + (\xi - \gamma_c)\,\frac{\psi U_c}{\theta f} - \eta^s\,\frac{\sigma + \xi_c}{c}$$

at each future node.

Proof. See Appendices E.3 through E.6. □

Corollary 7.2 (a frozen savings wedge tilts the carbon price).
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$*,*

$$\frac{\tau_t^q - SCC_t}{1 + \tau_t^q} = \eta_t^s \, \mathcal{I}_t,$$

and the deviations cannot all share one sign: $\eta_t^s \geq 0$ *at every node, or* $\eta_t^s \leq 0$ *at every node, forces* $\vartheta \equiv 0$ *and* $\eta^s \equiv 0$ *(Appendix E.2). If the schedule is not the virtual optimum and the dirty good is normal, the carbon price lies above the social cost of carbon at some nodes and below it at others.*

A two-period example signs the tilt. Let preferences separate the goods from labor and from each other, and let the schedule overtax savings, $\bar{\tau}_0^s > 0$ against a virtual wedge of zero. The frozen Euler equation front-loads consumption: $U_{c,0}$ must sit below $\beta R \, U_{c,1}$. The numéraire is the only good whose delivery moves its own marginal utility, so the planner overdelivers it early, underdelivers it late, and the dirty good absorbs the mirror image: $\eta_0^s > 0 > \eta_1^s$ (Appendix E.2), so the carbon price starts above the social cost of carbon and ends below it. A savings subsidy tilts the profile the other way, and the labor wedge tilts alongside through the first formula of Result 7.4. The reversal is not an artifact of two periods. The savings wedge restricts ratios of marginal utilities, so making the numéraire scarce at one date makes it abundant relative to its neighbors, and a deviation profile lying always on one side of the social cost of carbon would need the requirement to relax at every date at once, which the boundary conditions $\vartheta_0 = 0$ and $\eta_T^s = \vartheta_T U_{c,T}$ rule out. However badly the savings schedule is misset, it cannot justify a uniform carbon markdown or markup; it reshuffles the carbon price over the life cycle.

The contrast between the two exercises is instructive. A missing labor optimum moves the carbon price node by node, in one direction per node, because the labor constraint misallocates the numéraire across types within a date. A missing savings optimum tilts the carbon price across dates, in both directions, because the savings constraint misallocates the numéraire across dates within a life. The same two statistics price the leak in both cases, $\mathcal{I}_t$ on the commodity margin and $(\sigma + \xi_c)/c$ inside the social cost of carbon; only the multiplier changes. Retirement separates them most sharply: the labor-frozen distortions die with the labor margin, while the savings margin operates through retirement, where the virtual wedge is zero, so a nonzero frozen wedge keeps injecting error and retirees' carbon price deviates from the social cost of carbon. The Pigouvian-in-retirement result of Corollary 6.2 survives a frozen labor wedge and fails against a frozen savings wedge.

The intertemporal margin also admits the optimal-flat-level exercise, with a twist the static margins lack.

Result 7.5 (the optimal flat savings tax).
Let the schedule be flat, $\bar{\tau}_t^s(\theta) = \bar{\tau}_t^s$*, with its level chosen optimally at every date. The optimal*

level zeroes the U_c-weighted average of the forward multipliers,

$$\mathbb{E}_0[\vartheta_{t+1}\, U_{c,t}] = 0 \qquad \Longleftrightarrow \qquad \mathbb{E}_0[\eta^s_t] = \mathbb{E}_0[\vartheta_t\, U_{c,t}]\,,$$

and at $t = 0$, where nothing is inherited, $\mathbb{E}_0[\eta^s_0] = 0$.

Proof. See Appendix E.7. □

The condition prices only what the date-t rate controls. The rate moves the marginal-utility requirements passed forward, valued at ϑ_{t+1} and weighted by how hard the constraint binds through $U_{c,t}$; the requirements inherited from yesterday's rate it cannot touch. So the multiplier averages to zero only at birth, and afterward its target drifts with the inherited term $\vartheta_t U_{c,t}$: the level-versus-drift asymmetry of Lemma 7.2 survives optimization. Even at the optimal flat path the multiplier is generally nonzero node by node, so the carbon-price deviations of Corollary 7.2 do not vanish; the optimal level disciplines their average, not their profile. Flatness costs here what uniformity cost in Result 6.5: a single rate cannot track the cross-section, and the residual dispersion lands on the free margins.

7.3 A constrained commodity wedge

The last margin to freeze is the one legislatures actually freeze. Carbon prices are set by statute, capped after protests, or repealed outright, and none of these levels needs to coincide with the social cost of carbon. Section 5 asked what the carbon price should be; this subsection takes the carbon price as whatever legislation says and asks what the labor and savings wedges should do about it. The planner solves the problem of Section 5 with the commodity wedge held to an exogenous schedule $\bar{\tau}^q_t(\theta)$, node by node,

$$\frac{U_q\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} = 1 + \bar{\tau}^q_t(\theta) \equiv \bar{p}_t(\theta), \tag{5}$$

with multiplier $\eta^q(\theta) f_t(\theta|\theta_-)$: the constraint of Section 6, now with a schedule that is exogenous and may vary with the type. The leading case is a flat legislated price, $\bar{\tau}^q_t(\theta) = \bar{\tau}^q_t$, but the derivation allows any schedule. Like the labor constraint and unlike the savings one, (5) is static, so the multiplier is pinned by a static condition, and the regularity it needs is the strict quasi-concavity $\mathcal{D}_t < 0$ of Section 6.

Lemma 7.3 (the price of the frozen commodity wedge)**.**
At every history,

$$\eta^q_t = \bar{p}_t\; |s_{qq,t}| \left[(\gamma_{c,t} - \gamma_{q,t})\, \frac{\psi U_{c,t}}{\theta f_t} - \frac{\bar{\tau}^q_t - SCC_t}{1 + \bar{\tau}^q_t} \right],$$

where SCC_t is the shadow social cost of carbon of Result 7.6 below. The bracket is the virtual deviation the planner would set, $(\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f}$ by Result 5.1, minus the frozen one: $\eta_t^q > 0$ where the schedule underprices the dirty good and $\eta_t^q < 0$ where it overprices it, and the compensated substitution response converts the pricing error into the multiplier.

Proof. See Appendix C.4. □

The planner can no longer set the carbon price, but marginal damages do not leave the problem: the shadow SCC_t inside the Lemma is what routes the damages the frozen price fails to collect into the remaining wedges.

Result 7.6 (wedges under a frozen commodity wedge)**.**
At every history, the labor wedge and the savings adjustment are

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t\,\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^q\,\bar{p}_t\left[\mathcal{I}_t - \frac{\gamma_{c,t}-\gamma_{q,t}}{(1-\tau_t^y)\,y_t}\right], \qquad \chi_t^q \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q\,\bar{p}_t\,\mathcal{I}_t,$$

with the generalized inverse Euler equation holding in $U_{c,t}/(1-\chi_t^q)$, and the social cost of carbon keeps its discounted-damage form with the damage weight

$$1 + (\xi - \gamma_c)\,\frac{\psi U_c}{\theta f} + \eta^q\left[\bar{p}\,\frac{\sigma+\xi_c}{c} - \frac{\sigma_{cq}+\xi_q}{q}\right]$$

at each future node, where $\xi_{q,t} \equiv U_{Sq,t}\,q_t/U_{S,t}$ is the complementarity of the pollutant with the dirty good.

Proof. See Appendices C.5, C.6, and C.11. □

Corollary 7.3 (the income tax collects the carbon the carbon price misses)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrections collapse to the marginal budget share $m_{q,t}$ of Section 6 times the uncollected carbon price:

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t\,\frac{\psi U_{c,t}}{\theta f_t} + m_{q,t}\,\frac{SCC_t - \bar{\tau}_t^q}{1+\bar{\tau}_t^q}, \qquad \chi_t^q = \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - m_{q,t}\,\frac{SCC_t - \bar{\tau}_t^q}{1+\bar{\tau}_t^q}.$$

A carbon price frozen at the social cost of carbon is harmless: $\bar{\tau}_t^q = SCC_t$ gives $\eta_t^q = 0$ and returns every wedge to its Section 5 form. A carbon price frozen below it raises the labor wedge above the virtual level; one frozen above it lowers the labor wedge.

Proof. See Appendix C.5. □

The mechanism is embedded carbon. A marginal compensated dollar of earnings is spent, the fraction $m_{q,t}$ of it on the dirty good, and each of those units carries uncollected marginal damage

$(\mathrm{SCC}_t - \bar{\tau}_t^q)/(1 + \bar{\tau}_t^q)$. The income tax adds exactly that charge: it taxes the carbon content of marginal spending that the carbon price misses. The savings side applies the same charge intertemporally. Delivering consumption at date $t+1$ rather than t carries the uncollected damage of the extra dirty consumption it drags along, so the planner taxes savings where underpricing worsens over the life cycle and subsidizes it where pricing improves; a legislated price that stays flat while the social cost of carbon grows is precisely the worsening case. Both corrections persist into retirement, since the pricing error is a within-period object that never expires: like the frozen savings wedge and unlike the frozen labor wedge, a mispriced carbon tax distorts retirees' margins. This is the Mirrleesian counterpart of Hänsel et al. (2022), who find that a mandated uniform carbon tax reshapes the optimal income tax; here the reshaping has a closed form, and it is the marginal budget share times the pricing error.

Two connections complete the triptych. First, Section 6 is the special case of this subsection in which the frozen schedule is flat and its level is chosen optimally, and Result 6.2 is precisely the statement that the optimal level zeroes the average multiplier, $\mathbb{E}_0[\eta_t^q] = 0$ date by date. Second, the section's message is symmetric. Freezing any wedge at its virtual optimum is harmless; freezing it anywhere else sends the error to whichever margins remain free, with income effects as the conduit. A labor error moves the carbon price node by node; a savings error tilts the carbon price across dates; a carbon error raises or lowers the labor wedge and drifts the savings wedge. The separation of Proposition 5.1 survives restricting the form of the corrective instrument, as Section 6 showed; what it cannot survive is error, in any of the three.

7.4 A Mirrlees-Ramsey bridge

The section's exercises assemble into a bridge between the two traditions the literature review set against each other. Each margin now admits three readings. Left free, it carries its Mirrleesian wedge. Frozen flat with the level chosen optimally, it is a linear instrument of the Ramsey kind, and the optimality condition is the same on every margin: the multiplier on the wedge constraint averages to zero across the cross-section,

$$\mathbb{E}_0[\eta_t^q] = 0, \qquad \mathbb{E}_0[\eta_t] = 0, \qquad \mathbb{E}_0[\vartheta_{t+1}\, U_{c,t}] = 0,$$

for the commodity price (Result 6.2), the labor rate (Result 7.3), and the savings rate (Result 7.5). Frozen anywhere else, the margin is at the third best, and the multiplier's profile prices the error. Margin by margin, the optimal linear instrument is the zero-average point of the third-best family, and the second best of Section 5 is the point where every multiplier vanishes identically.

8 Quantitative analysis

How large is the Mirrleesian correction to the carbon price, and how much welfare does it deliver over a flat carbon tax? I answer both by solving the model's optimal mechanism globally, calibrated to US data, rather than evaluating the wedge formula at borrowed sufficient statistics. The solver extends the dual-costate recursion of Farhi and Werning (2013) and Golosov et al. (2016) to two consumption goods, a clean numéraire and a polluting energy good, and prices the externality as a constant marginal damage e entering the planner's resource constraint, so the social producer price of energy is $1 + e$. This is the single-cohort reading of the social cost of carbon in Section 5: for a cohort too small to move the stock, the discounted damage weights Φ_j collapse to one constant per-unit cost, no pollution-stock state is carried, and the Pigouvian benchmark is $\tau^q = \text{SCC} = e$. The stock dynamics are a separate exercise. What the solved mechanism settles is the object a formula evaluation cannot: the *level* and the *incidence* of the optimal commodity wedge, and the welfare cost of implementing it with an anonymous instrument.

8.1 Model and calibration

The life-cycle block is Farhi and Werning (2013)'s, replicated and validated before the two-good extension. Agents live sixty years, work for forty and retire for twenty; log-skills follow a geometric random walk with innovation variance $\hat{\sigma}^2 = 0.0095$; $\beta = 1/R = 0.95$, so $\beta R = 1$; skills are private information and the incentive constraints are handled by the first-order approach. Each period the agent consumes a clean numéraire and an energy good (motor fuel, electricity, gas, fuel oil), both with producer price one. Period utility is

$$U(c, q, l) = \frac{1}{\rho} \log\left[(1 - b_q)\, c^\rho + b_q\, e^{-\chi l^\alpha}\, (q - \bar{q})^\rho\right] - \frac{l^\alpha}{\alpha}, \qquad \rho \equiv 1 - \frac{1}{\varsigma},$$

a CES aggregate over the clean good and energy net of subsistence $\bar{q}$, inside the log shell that preserves Farhi and Werning (2013)'s balanced-growth homogeneity, with isoelastic labor disutility of curvature $\alpha = 3$ (Frisch elasticity one half). ς is the elasticity of substitution between the goods, b_q the energy weight, and $\bar{q} = 0$ in the homothetic benchmark; $\bar{q} > 0$ is the Stone-Geary necessity variant. The twist $e^{-\chi l^\alpha}$ makes energy a relative leisure complement: its utility weight falls with labor effort, so $U_{ql} \neq 0$ with the sign, in the notation of Section 4, that makes the complementarity gap $\gamma_c - \gamma_q$ positive. This is the empirically relevant case (West and Williams III (2007); Christiansen (1984)), and the one that pushes the optimal carbon tax above Pigouvian.

Two solvers deliver the results. The *second best* assigns full consumption bundles, so the commodity wedge is unrestricted; this solver requires homothetic preferences. *Linear commodity taxation* layers an anonymous rate τ^q (flat or age-dependent) on the optimal nonlinear income and savings mechanism, and reduces exactly to a one-good problem with reshaped labor disutility, valid with subsistence and hence usable for the necessity calibration. The two solvers agree where they must: at $\gamma_c = \gamma_q$ the second best sets $\tau^q = e$ for every history and every age, the dynamic Atkinson-Stiglitz result, and it coincides with a flat Pigouvian tax to solver precision, returning the one-good initial condition $\hat{\lambda}_1 = 0.8797$. The incentive compatibility of the first-order-approach solution is verified ex post on the simulated benchmark.

The two-good calibration is disciplined by four US moments (Table 2). The energy expenditure share is 7% (Consumer Expenditure Survey, 2023). The income elasticity of energy is 0.45, below one because energy is a necessity, from residential-energy demand studies; the homothetic benchmark raises it to one, and the Stone-Geary variant keeps it at 0.45. The own-price elasticity is 0.4, from meta-analyses of energy demand; at a 7% share it pins the substitution elasticity $\varsigma \approx 0.4$. The leisure complementarity χ is set so that, with the externality off, the second-best energy wedge at mid-career is about 10% ad valorem: West and Williams III (2007)'s zero-externality gasoline tax, tempered for the household-energy composite, whose labor cross-effects are weaker than motor fuel's. The marginal damage is $e = 0.494$: the EPA's central social cost of carbon of \$190 per ton (U.S. Environmental Protection Agency 2023) times the carbon intensity of household energy spending, 0.0026 tons per producer dollar. Because $p_2 = 1$, the Pigouvian wedge $\tau^q = e$ is a 49% ad valorem tax on energy.

Table 2: Calibration

parameter	value	moment or source
life-cycle block (Farhi and Werning 2013)		
discount factor $\beta = 1/R$	0.95	FW13 ($\beta R = 1$)
labor-disutility curvature α	3	Frisch elasticity 0.5
log-skill innovation variance $\hat{\sigma}^2$	0.0095	US male wages, via FW13
working / retirement periods	40/20	FW13
two-good block		
energy expenditure share	0.07	Consumer Expenditure Survey 2023
income elasticity of energy	0.45	residential-energy demand (necessity)
own-price elasticity of energy	0.4	energy-demand meta-analyses
leisure complementarity χ	0.25	mid-career zero-externality wedge $\approx$ 10% (West and Williams III 2007)
marginal damage e	0.494	\$190/tCO_2 (U.S. Environmental Protection Agency 2023) $\times$ 0.0026 tCO_2/\$

8.2 The optimal carbon wedge over the life cycle

The optimal wedge is the social cost of carbon plus a small, rising premium (Table 3, Figure 1). At labor-market entry the labor wedge is near zero, there is no incentive motive yet, and the carbon wedge sits at the Pigouvian level $e = 0.494$. It then rises concavely to about 0.63

near retirement. The non-Pigouvian premium, the markup over the Pigouvian gross price, tops out around 9%. So the optimal carbon tax exceeds the social cost of carbon, in the direction a leisure complement predicts, but by less than a tenth over most of the life cycle.

The premium is a scaled labor wedge. Its first-order value is $\chi\, l^{\alpha}\, \tau^{y}/(1-\tau^{y})$, and this identity tracks the solved premium within about 10% at every age (last two rows of Table 3). The non-Pigouvian part of the carbon wedge therefore inherits the labor wedge's level and its life-cycle rise, which is the proportionality result of Section 4 made quantitative. The wedge is also genuinely history-dependent: its cross-sectional standard deviation fans out from 0.1% early in life to 5.5% near retirement, mirroring the spreading of the labor wedge as skill histories diverge.

The income-side wedges are essentially undisturbed by the second good. The labor wedge runs from 0.02 to 0.36 over the working life, against 0.02 to 0.37 in the one-good economy, and the savings wedge falls from 0.56% to 0.02%, the Farhi and Werning (2013) pattern intact. The commodity margin adds a corrective and screening instrument without reshaping the labor and savings margins, which is the separation showing up in the solved policies. Under the wedge the mean energy share falls from the calibrated 7% to about 6% at producer prices, the households priced off energy at the calibrated elasticity.

Table 3: Cross-sectional mean wedges over the working life (benchmark, second best)

working year t	1	10	20	30	40
labor wedge τ^{y}	0.021	0.167	0.272	0.336	0.363
carbon wedge τ^{q}	0.504	0.569	0.606	0.624	0.632
s.d. across histories	0.001	0.005	0.014	0.029	0.055
non-Pigouvian premium	0.007	0.050	0.075	0.087	0.092
identity $\chi l^{\alpha}\tau^{y}/(1-\tau^{y})$	0.006	0.047	0.069	0.079	0.083

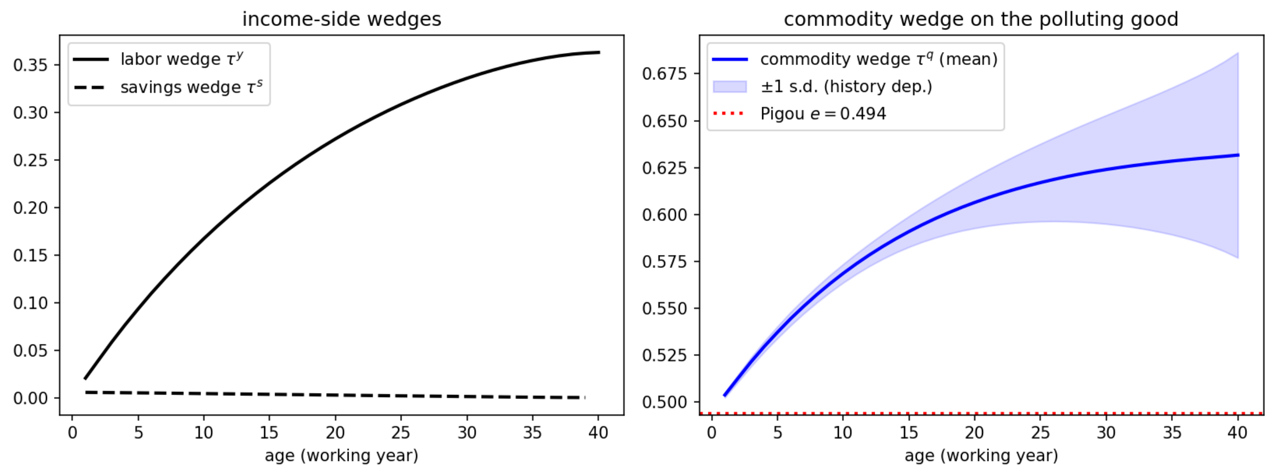

Figure 1: Income-side wedges (left) and the commodity wedge on energy (right), benchmark calibration. The band is ± 1 cross-sectional standard deviation across histories; the dotted line is the Pigouvian level $\tau^q = e = 0.494$.

8.3 The optimal wedge is progressive in skill

The optimal carbon wedge is progressive in skill (Figure 2). Binned by current productivity, the wedge is indistinguishable across quintiles early in life, when every history looks alike, and it fans out from mid-career: by the last working year it reaches about 70% for the top skill quintile against 57% for the bottom, a 13-point spread. High current productivity signals a history whose incentive constraints are expensive, and the leisure-complement margin is the planner's lever on exactly those histories, so the screening value of the commodity instrument concentrates on high-skill agents. Figure 2 is a schedule of marginal wedges by skill, not a map of the burden each household bears. That burden is set jointly by the nonlinear income tax and transfers alongside the wedge, so the schedule's progressivity speaks to the corrective instrument itself, not to the full incidence of carbon pricing.

Two readings follow. The optimal carbon wedge is not regressive: the planner taxes the energy of high-skill households more, not less, so the redistributive worry about carbon pricing is misplaced at the level of the optimal type-contingent wedge. The regressivity of real carbon taxes comes from elsewhere, from the flat-rate restriction, since an anonymous point-of-sale tax cannot condition on skill, and, in the necessity calibration, from energy's falling budget share. That is also why the flat rungs of the welfare ladder below leave money on the table.

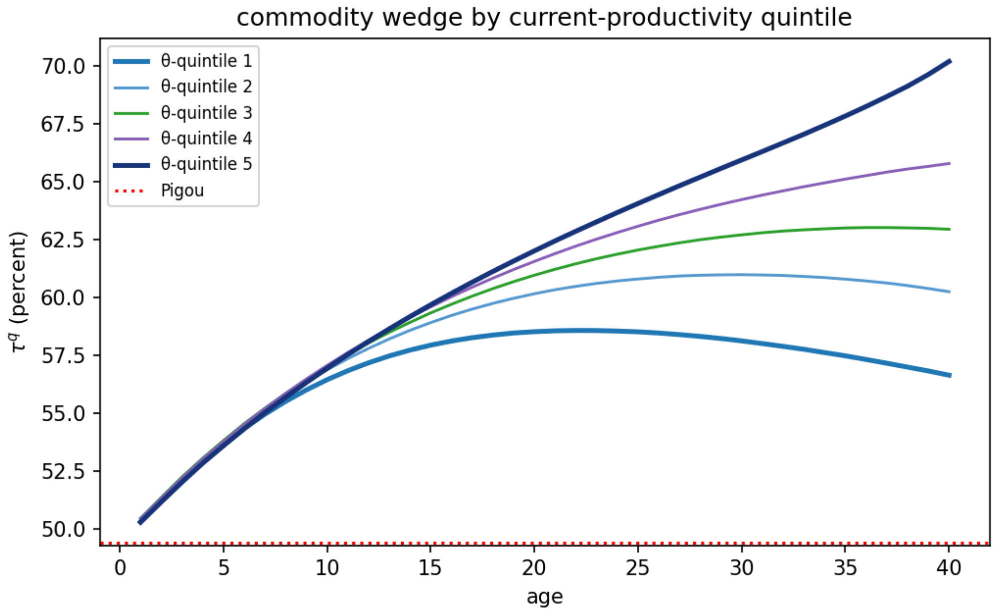

Figure 2: The optimal commodity wedge τ^q (%) by current-skill quintile and age, benchmark calibration.

8.4 The welfare value of carbon pricing and the cost of simpler rules

Table 4 is the welfare accounting: consumption-equivalent gains over having no carbon price, under the optimal nonlinear income and savings mechanism throughout, computed with common random numbers so that cross-regime differences net out simulation noise. Every regime pays for its own emissions in the planner's budget, so no rung is flattered by unpriced damages; the numerical resolution is about 0.002 percentage points, and differences below it are noise. Six findings, in order of size.

The value of a carbon price is overwhelmingly Pigouvian. At the EPA social cost of carbon, having any carbon price is worth 0.31% of consumption, and a flat rate at the social cost of carbon alone captures 98% of the full second-best value while cutting emissions by 13.5%. For scale, the entire insurance value of Farhi and Werning (2013)'s optimal income-tax apparatus over laissez-faire is 1.56% of consumption: the carbon instrument is worth about a fifth of that at this social cost of carbon.

The optimal flat rate exceeds the social cost of carbon by about 14%, $\tau^{q*} = 0.56$ against $e = 0.494$, the flat reflection of the Corlett-Hague premium. But the welfare gain from that uplift over pricing at the social cost of carbon is 0.003%: the objective is very flat in the rate, so "price at the social cost of carbon" is a near-optimal rule of thumb even with nonseparable preferences.

Sophistication buys almost nothing beyond the level. Making the rate age-dependent adds about 0.002 percentage points over the optimal flat rate, and full history dependence about 0.001 more, both at the numerical resolution. The premium varies over life and across histories by a few points of a 6% budget share with a substitution elasticity of 0.4, so the deadweight loss from mistracking it is second-order squared.

The pure Corlett-Hague motive is tiny. With the externality off, the entire commodity-wedge apparatus of the second best is worth 0.0036% of consumption, and its optimal flat rate is 5%, materially positive as in West and Williams III (2007) but negligible in welfare. The flat rate sits below the 10% mid-career calibration target because it averages a premium that starts the working life near zero. Non-separability matters for the *level* of the optimal carbon tax, the 14% uplift, not as a stand-alone reason to tax energy.

Necessity halves the value and the abatement. Under the Stone-Geary calibration, in which energy's budget share falls with income, the value of the flat carbon tax falls to 0.15% and its abatement to 6%, because the subsistence part of energy does not respond to price. The optimal flat rate is essentially unchanged. This is the calibration in which a flat carbon tax is regressive, yet the mechanism absorbs the incidence through the income side, which is why the flat instrument loses so little welfare despite its incidence.

The separation placebo. With the leisure complementarity switched off ($\gamma_c = \gamma_q$), the second best and the flat Pigouvian tax coincide to five decimal places in welfare and in emissions: dynamic Atkinson-Stiglitz, and the exact knife-edge of Proposition 5.1, reproduced through two structurally different solvers.

Table 4: Welfare value of carbon pricing under progressively simpler rules (consumption-equivalent gain over $\tau^q = 0$, % of consumption)

	benchmark ($e = 0.494$)	Stone-Geary (necessity)	$\gamma_c = \gamma_q$ (placebo)	no externality ($e = 0$)
second best (nonlinear τ^q)	0.315	—	0.252	0.0036
age-dependent linear $\tau^q(t)$	0.314	—	—	0.0038
flat τ^{q*} (optimized)	0.313	0.153	—	0.0026
optimal flat rate τ^{q*}	0.56	0.56	—	0.051
flat Pigouvian $\tau^q = e$	0.309	0.151	0.252	—
emissions vs $\tau^q = 0$: second best	−15.0%	—	−13.5%	−1.8%
flat Pigouvian	−13.5%	−6.3%	−13.5%	−1.8%

Note: welfare entries are in % of consumption; the optimal-flat-rate row is an ad-valorem rate, and the final two rows report emissions changes relative to $\tau^q = 0$. Differences below the 0.002-point numerical resolution, including the apparent ranking of the age-dependent rate above the second best in the no-externality column, are simulation noise. Dashes mark three cases. *Not computed*: the second best requires homothetic preferences, so it and the age-dependent rung are blank under the non-homothetic Stone-Geary calibration. *Degenerate*: under the $\gamma_c = \gamma_q$ placebo the second best already coincides with the flat Pigouvian tax, so the intermediate rungs add nothing. *Not applicable*: with no externality ($e = 0$) there is no Pigouvian benchmark.

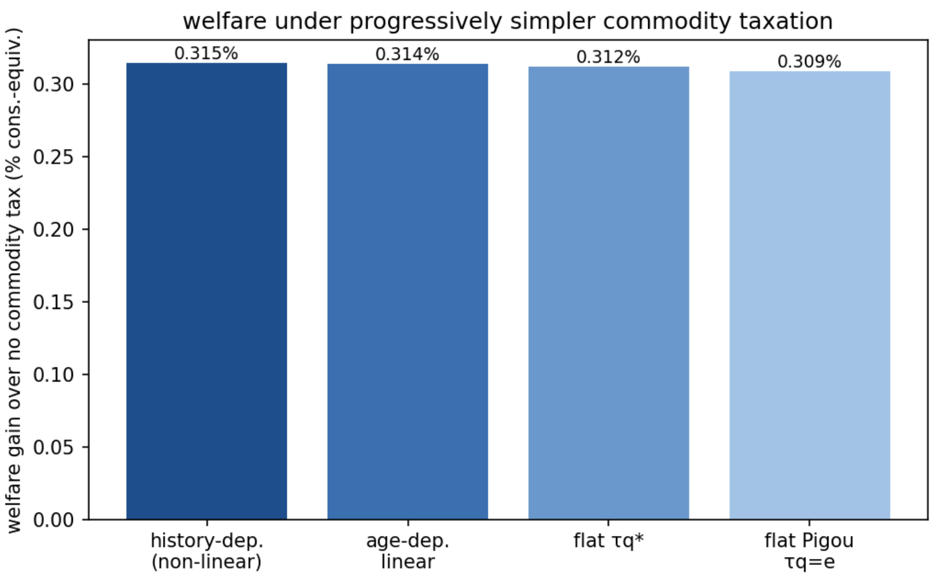

Figure 3: Consumption-equivalent welfare gain over no carbon price, benchmark calibration, under progressively simpler carbon-pricing rules.

8.5 Reading and limitations

The practical hierarchy for carbon policy in a Mirrleesian world is clean. Get the carbon price to the social cost of carbon, worth 0.31% of consumption; nudge it up by the premium for leisure complementarity, worth a further 0.003%; and spend the sophistication budget on the income tax, where Farhi and Werning (2013)'s own accounting shows the returns to age- and history-dependence are two orders of magnitude larger. The redistributive value embedded in the carbon tax is small, which is the separation principle in welfare terms: the income tax and the climate dividend, not the carbon price, should carry the distribution. This agrees with Douenne et al. (2026)'s near-Pigouvian conclusion, reached here from the Mirrleesian rather than the Ramsey side, and it answers the distributional concern that opens the paper. Regressive carbon pricing and the backlash it provokes are best met by adjusting the income tax, not by tilting, means-testing, or weakening the carbon price.

Four limitations bound the numbers. The externality is a constant marginal damage, so the stock dynamics of Section 5, which move the *level* of the tax rather than the wedge structure measured here, are a separate exercise. The second best requires homothetic preferences, so the necessity calibration is read off the exact linear-tax reduction rather than the full nonlinear mechanism. The log consumption index is required by the balanced-growth homogeneity the solver relies on. And the energy good carries a single carbon intensity, so within-energy heterogeneity is outside the model. None of these bears on the three results: the sign and proportionality of the premium, its progressivity in skill, and the near-sufficiency of a flat Pigouvian price are structural.

9 Conclusion

A pollution externality placed inside a dynamic Mirrlees economy leaves the corrective and redistributive tasks separate. When preferences are weakly separable, the carbon wedge is exactly the social cost of carbon and the labor and savings wedges are unchanged, so the below-Pigouvian markdown that the Ramsey tradition blames on fiscal distortion does not survive an optimal nonlinear income tax. Only nonseparability bends the carbon wedge away from the social cost of carbon, and it does so in the direction the good's complementarity with labor dictates.

Real carbon taxes are anonymous linear prices, not the history-dependent wedge the theory prescribes. Under that restriction the two-good problem collapses to a one-good problem in expenditure; the optimal uniform rate is a substitution-weighted average of the wedges it replaces, and its welfare cost is their dispersion, which vanishes under weak separability. With the externality the uniform price splits additively into the social cost of carbon and a screening term, so the restriction binds only the redistributive part and never the Pigouvian one.

Carbon pricing is resisted as a burden on the poor, and governments answer by capping it, tilting it, or outright repealing it. This paper suggests a different response: once the income tax is set optimally, the carbon price has no real redistributive role and should simply be set at the social cost of carbon.

This reading carries one caveat. The separation is a property of the jointly optimal tax system, and presumes the labor and savings wedges are set optimally alongside the carbon price. When the labor wedge is held above or below its optimum, part of the redistribution it fails to carry falls back on the carbon price: Section 7 signs the deviation, below the social cost of carbon where the frozen schedule overtaxes labor and above it where the schedule undertaxes. A frozen savings wedge works differently: its error can only tilt the carbon-price profile, above the social cost of carbon at some dates and below it at others. And the flow runs both ways: a carbon price legislated below the social cost of carbon raises the optimal labor wedge by the marginal budget share of the dirty good times the uncollected carbon price, the income tax collecting the carbon the carbon price misses. The optimal linear instruments of the Ramsey tradition emerge inside the same family, as the levels at which the constraint multipliers average to zero.

Three extensions stand out. First, the paper characterizes wedges, not their implementation; how to replicate the optimal commodity wedge with real instruments is open. Second, the model follows a single generation, but climate change is intergenerational. The natural next step is an overlapping-generations extension, in which the single-generation model becomes the component problem of an infinite-horizon planner. The fixed point between the carbon-price path and the per-cohort mechanism-design problems would then endogenize the savings-wedge paths that Section 8 takes from calibrated optima. Third, outside the environmental context, the optimal commodity wedge is a time-varying object, which raises the question of age-dependent commodity taxation. Age-dependent commodity taxes may sound fanciful, yet age-based pricing of some goods and services, and discounts for low-income consumers, already implements a time-varying, income-based commodity wedge.

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A Optimal wedge derivation without externalities

Because $U(c, q, l)$ can be written as $U(c, q, y/\theta)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian of the recursive problem is

$$
\begin{aligned}
\mathcal{H} = &\left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta)\right] f \\
&+ \psi \left[-U_l(c, q, y/\theta)\frac{l}{\theta} + \beta w_2\right] \\
&- \lambda_1 u(\theta)\bar{\alpha}_t f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi \left[u - U(c, q, y/\theta) - \beta w\right],
\end{aligned}
$$

Differentiating yields the following first-order conditions

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql}\frac{l}{\theta} = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l = \frac{1}{\theta}\psi \left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta.
\end{aligned}
$$

A.1 Labor wedge

Dividing the first-order condition for c by U_c and using $\gamma_c \equiv U_{cl} l/U_c$,

$$
\phi = \frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c. \tag{6}
$$

From the first-order condition for l, and since $U_{ll} l + U_l = U_l(1 + \epsilon)$ with $\epsilon \equiv U_{ll} l/U_l$,

$$-\theta f - \phi U_l = \frac{\psi}{\theta} U_l (1+\epsilon) \implies \phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon).$$

Equating the two expressions for ϕ and collecting the terms in ψ,

$$\frac{f}{U_c} + \frac{\theta f}{U_l} = -\frac{\psi}{\theta}(1+\epsilon-\gamma_c).$$

The labor wedge is $1-\tau^y = -U_l/(\theta U_c)$, so $\theta U_c/U_l = -1/(1-\tau^y)$, making the left-hand side $\frac{f}{U_c}\left(1-\frac{1}{1-\tau^y}\right) = -\frac{f}{U_c}\frac{\tau^y}{1-\tau^y}$. Substituting and rearranging,

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{\psi U_c}{\theta f} = A_t(\theta)\,\frac{\psi U_c}{\theta f} \tag{7}$$

Next, substitute ϕ from (6) into $\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi$,

$$\dot{\psi} - \frac{\gamma_c}{\theta}\psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}.$$

This is a linear first-order ordinary differential equation. Multiplying by the integrating factor $\exp\left[-\int^{\theta}\gamma_c(\tilde{x})\,d\tilde{x}/\tilde{x}\right]$ and integrating from θ to ∞ with the boundary condition $\psi(\infty)=0$,

$$\psi(\theta) = \int_{\theta}^{\infty} \exp\left[-\int_{\theta}^{x}\gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx. \tag{8}$$

The remaining boundary condition $\psi(0)=0$ pins down λ_1,

$$\lambda_{1,t} = \frac{\int_0^{\infty}\exp\left[-\int_0^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{1}{U_{c,t}(x)}f_t(x) + \lambda_{2,t}f_{2,t}(x)\right)dx}{\int_0^{\infty}\exp\left[-\int_0^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\bar{\alpha}_t(x)f_t(x)dx}. \tag{9}$$

Substituting (8) into (7),

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{U_c(\theta)}{\theta f(\theta)}\int_{\theta}^{\infty}\exp\left[-\int_{\theta}^{x}\gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx.$$

I split the integral into the $[f/U_c - \lambda_1 \bar{\alpha}_t f]$ part, which yields the intratemporal term $A_t B_t C_t$, and the $\lambda_2 f_2$ part, which yields the intertemporal term. For the first part, carry $U_c(\theta)/U_c(x)$ into the integrand. Along the optimal allocation $U_{c,t}$ depends on the type through $c(\theta)$, $q(\theta)$, and $l(\theta) = y(\theta)/\theta$, so

$$\frac{d}{d\theta} \ln U_{c,t} = \frac{U_{cc}\dot{c} + U_{cq}\dot{q} + U_{cl}\dot{l}}{U_c} = -\sigma_t \frac{\dot{c}}{c} - \sigma_{cq,t}\frac{\dot{q}}{q} + \gamma_c \frac{\dot{l}}{l}.$$

Since $l = y/\theta$ gives $\dot{l}/l = \dot{y}/y - 1/\theta$, integrating from θ to x and multiplying by the integrating factor gives the identity

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] = \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_c(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right], \quad (10)$$

in which the $\gamma_c/\tilde{x}$ term from $\dot{l}/l$ cancels the integrating factor. The cross-curvature $\sigma_{cq,t}$ enters because, under nonseparability, extracting resources from a higher type shifts the marginal utility of the numéraire through the adjustment of both goods. The first part therefore equals $A_t B_t C_t$. For the second part, two preliminaries pin down $\lambda_{2,t}$. The first-order condition for w_2 rearranges to $\partial V_{t+1}/\partial w_2 = -\beta R\,\psi/f$, and the envelope theorem applied to the period-$(t+1)$ problem gives $\partial V_{t+1}/\partial \hat{w}_2 = -\lambda_{2,t+1}$, the multiplier with which threat keeping enters the Lagrangian. Equating the two expressions and shifting the date back one period,

$$\lambda_{2,t}(\theta^{t-1}) = \beta R\, \frac{\psi_{t-1}}{f_{t-1}}.$$

Next, evaluating (7) at $t-1$ and solving for the costate,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,\frac{\theta_{t-1}}{U_{c,t-1}\,A_{t-1}}.$$

The $\lambda_2 f_2$ block of the wedge is $\lambda_{2,t}$, a constant in the current type, times the same outer structure as the first part,

$$A_t(\theta)\,\frac{U_{c,t}(\theta)}{\theta f_t(\theta)}\,\lambda_{2,t}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\,dx.$$

Substituting the two preliminaries and collecting the ratios,

$$= \beta R \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\,dx}{\theta f_t(\theta)} = \beta R \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} D_t(\theta).$$

Collecting both parts,

$$\frac{\tau^y}{1-\tau^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} D_t(\theta),$$

$$\begin{aligned}
A_t(\theta) &= 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \\
B_t(\theta) &= \frac{1-F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right] \left(1 - \lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)\right) \frac{f_t(x)dx}{1-F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}.
\end{aligned}$$

This is Golosov et al. (2016)'s labor-wedge decomposition, with the cross-curvature $\sigma_{cq,t}$ in C_t.

A.2 Savings wedge

The savings wedge follows from the same first-order conditions. Define

$$\chi_t(\theta^t) \equiv \frac{\gamma_{c,t}}{1+\epsilon_t - \gamma_{c,t}} \frac{\tau^y_t}{1-\tau^y_t},$$

Using $\frac{\tau^y_t}{1-\tau^y_t} = \frac{\psi U_c}{\theta f}(1+\epsilon_t-\gamma_{c,t})$, $\phi = f/U_c - \psi\gamma_c/\theta$ from (6) can be rewritten as follows:

$$\frac{\phi}{f} = \frac{1}{U_c}\left(1 - \gamma_c \psi U_c/(\theta f)\right) = \frac{1-\chi_t}{U_{c,t}}. \tag{11}$$

χ_t is positive when the numéraire is complementary with labor ($U_{cl} > 0$) and labor is taxed, and is zero when preferences are separable in the numéraire against labor ($\gamma_c = 0$), or when the labor margin is undistorted. The first-order condition for w and the envelope condition $\partial V_t/\partial \hat{w} = \lambda_{1,t}$ one period ahead give

$$\lambda_{1,t+1}(\theta^t) = \beta R\frac{\phi(\theta)}{f(\theta)} = \beta R\frac{1-\chi_t}{U_{c,t}}. \tag{12}$$

A second expression for $\lambda_{1,t}$ comes from integrating the costate equation over the whole type space. With $\psi(0) = \psi(\infty) = 0$,

$$0 = \int_0^\infty \dot{\psi}\, d\theta = \lambda_{1,t}\int_0^\infty \bar{\alpha}_t f\, d\theta - \lambda_{2,t}\int_0^\infty f_2\, d\theta - \int_0^\infty \phi\, d\theta.$$

The Pareto weights are normalized so that $\int \bar{\alpha}_t f\, d\theta = 1$, and $\int f_t(\theta|\theta_-)\, d\theta = 1$ for every θ_- implies $\int f_{2,t}\, d\theta = 0$. Hence $\lambda_{1,t} = \int_0^\infty \phi\, d\theta$, which by (11) is

$$\lambda_{1,t} = \mathbb{E}_{t-1}\left[\frac{1-\chi_t}{U_{c,t}}\right], \tag{13}$$

where $\mathbb{E}_{t-1}$ integrates over θ_t conditional on θ_{t-1}: the multiplier on promise keeping is set before the current draw is known. Evaluating (13) at $t+1$ and equating with (12),

$$\frac{1-\chi_t}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}}{U_{c,t+1}}\right].$$

Defining the incentive-adjusted marginal utility $\widehat{U}_{c,t} \equiv U_{c,t}/(1-\chi_t)$,

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right].$$

This is the inverse Euler equation in the adjusted marginal utility $\widehat{U}_c$: delivering a util through the numéraire costs $1/U_{c,t}$ in resources, and when $U_{cl} > 0$ the same unit lowers the information rent $-U_l\, l/\theta$, an incentive gain worth $\chi_t/U_{c,t}$, so the planner equalizes the present value of the net cost $(1-\chi_t)/U_{c,t}$. Section 4 maps this condition into the generalized inverse Euler equation of Hellwig (2021).

A.3 Commodity wedge

The commodity wedge uses the first-order conditions for c and q of the preamble. There are different equivalent ways to express it, but I will seek the familiar labor wedge $ABCD$ form. First, because the first-order conditions for each good share the same ϕ,

$$\frac{f_t}{U_q} - \psi\frac{U_{ql}}{U_q}\frac{l}{\theta} = \frac{f_t}{U_c} - \psi\frac{U_{cl}}{U_c}\frac{l}{\theta}$$

Cross-multiplying by U_q and U_c and using $\gamma_c \equiv \frac{U_{cl}\, l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}\, l}{U_q}$,

$$U_c\left(f - \frac{\psi}{\theta}\gamma_q U_q\right) = U_q\left(f - \frac{\psi}{\theta}\gamma_c U_c\right)$$

Dividing by $U_q f$ and rearranging,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f}(\gamma_q - \gamma_c)$$

Note that, by definition, $1 + \tau^q = \frac{U_q}{U_c}$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1+\tau^q}$. Multiplying through by -1,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$$

The prefactor $\psi U_c/(\theta f)$ is the same object that drives the labor wedge. Solving (7) for it and substituting the $ABCD$ decomposition of Appendix A.1,

$$\frac{\psi U_c}{\theta f} = \frac{1}{A_t}\frac{\tau^y}{1-\tau^y} = B_t C_t + \beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,\frac{D_t}{A_t}.$$

Multiplying by the complementarity gap $\gamma_{c,t} - \gamma_{q,t}$,

$$\frac{\tau^q}{1+\tau^q} = (\gamma_{c,t} - \gamma_{q,t})\left[B_t C_t + \beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,\frac{D_t}{A_t}\right],$$

with A_t, B_t, C_t, D_t as in Appendix A.1. Alternatively, multiplying $\frac{\psi U_c}{\theta f} = \frac{1}{A_t}\frac{\tau^y}{1-\tau^y}$ by the gap directly,

$$\frac{\tau^q}{1+\tau^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}}\frac{\tau^y_t}{1-\tau^y_t}$$

The commodity wedge inherits every dynamic feature of the labor wedge: its shape is that of the labor wedge, scaled by the elasticity term.

B Optimal wedge derivation with externalities

B.1 Commodity wedge

The Hamiltonian of this problem is below. As in Section 5, the fourth argument of V_{t+1} is the updated emission state $(Q_{t-s+1}, \ldots, Q_t)$, written as a single symbol, and μ is the multiplier on the date-t accumulation constraint.

$$
\begin{aligned}
\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, S)\right] f \\
& + \psi \left[-U_l(c, q, y/\theta, S)\frac{l}{\theta} + \beta w_2\right] \\
& - \lambda_1 u(\theta)\bar{\alpha}_t f_t \\
& + \lambda_2 u(\theta) f_2 \\
& + \phi \left[u - U(c, q, y/\theta, S) - \beta w\right] \\
& - \mu \left[S - S(\ldots, Q)\right]
\end{aligned}
$$

This yields the following first-order conditions.

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql}\frac{l}{\theta} + \left[\mu S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q}\right] f = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l = \frac{1}{\theta}\psi \left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S\right) d\theta = \mu
\end{aligned}
$$

Here $S_q \equiv \partial S/\partial Q$ is the marginal effect of current aggregate emissions on the current stock, and the density f in the first-order condition for q comes from differentiating the aggregate $Q = \int q(\theta) f(\theta)\, d\theta$, which enters both the current accumulation constraint and, as the newest entry of the emission state, the continuation value through $\partial V_{t+1}/\partial Q \equiv \partial V_{t+1}/\partial Q_t$. The current stock itself is not a state of the continuation problem, which is why no V term appears in the first-order condition for S. Define the full shadow price of current emissions,

$$\nu_t \equiv \mu_t S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q}. \tag{14}$$

As in the previous proof,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Because c and q share the same ϕ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Multiplying by U_q and U_c,

$$U_c\left(f - \psi U_{ql}\frac{l}{\theta} + \nu_t f\right) = U_q\left(f - \psi U_{cl}\frac{l}{\theta}\right)$$

Using $\gamma_c \equiv \frac{U_{cl}\,l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}\,l}{U_q}$, or, equivalently, $\gamma_c U_c = U_{cl}\,l$ and $\gamma_q U_q = U_{ql}\,l$,

$$U_c\left(f - \frac{\psi}{\theta}\gamma_q U_q + \nu_t f\right) = U_q\left(f - \frac{\psi}{\theta}\gamma_c U_c\right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta}U_c U_q(\gamma_q - \gamma_c) - \nu_t U_c f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f}(\gamma_q - \gamma_c) - \nu_t \frac{U_c}{U_q}$$

Note that, by definition, $1 + \tau^q = U_q/U_c$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1+\tau^q}$. Multiplying through by -1,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \nu_t \frac{U_c}{U_q}$$

Since $\frac{U_c}{U_q} = \frac{1}{1+\tau^q}$,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \frac{\nu_t}{1+\tau^q}$$

Rearranging,

$$\frac{\tau^q - \nu_t}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$$

The right-hand side is exactly that of the no-pollution primitive condition $\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$ of Appendix A.3, with τ^q replaced by the deviation $\tau^q - \nu_t$. The first-order conditions for c and l and the costate equation for ψ are untouched by the externality, so the integrating-factor steps there carry over verbatim and give

$$\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right],$$

with A_t, B_t, C_t, D_t the labor-wedge terms of Appendix A.1 (the cross-curvature $\sigma_{cq,t}$ included in C_t). The externality leaves this block untouched and enters only by replacing τ^q with the deviation $\tau^q - \nu_t$ on the left.

Now consider ν_t. Its first piece is the date-t stock multiplier μ_t, the marginal damage of a unit of the current stock. Substituting ϕ from the first-order condition for c into the first-order condition for S, then dividing and multiplying by f,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \left[\frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c} \right] U_S \right) d\theta = \int_0^\infty \left(-\frac{\psi}{\theta f} U_{Sl} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l \right] \frac{U_S}{U_c} \right) f \, d\theta.$$

Rewrite $\psi/\theta f$ by noting that

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{U_c}\frac{1}{1+\epsilon-\gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{Sl}l}{U_S}$,

$$\begin{aligned}
\mu_t &= \int \left(-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{1+\epsilon-\gamma_c}\frac{U_{Sl}l}{U_c} - \left[1-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{1+\epsilon-\gamma_c}\frac{U_{cl}l}{U_c}\right]\frac{U_S}{U_c}\right) f\, d\theta \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{1+\epsilon-\gamma_c}\frac{U_{Sl}l}{U_c}\frac{U_S}{U_S} - \left[1-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma_c}{1+\epsilon-\gamma_c}\right]\frac{U_S}{U_c}\right) f\, d\theta \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\xi}{1+\epsilon-\gamma_c}\frac{U_S}{U_c} - \left[1-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma_c}{1+\epsilon-\gamma_c}\right]\frac{U_S}{U_c}\right) f\, d\theta \\
&= -\int \left[1+\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\xi-\gamma_c}{1+\epsilon-\gamma_c}\right]\frac{U_S}{U_c}\, f\, d\theta.
\end{aligned}$$

The second piece of ν_t prices the echo of current emissions in future stocks. By the envelope theorem applied to the period-$(t+1)$ problem, Q_t enters continuation costs only through the accumulation constraints of the next s periods: $S_{t+j} = S(Q_{t+j-s}, ..., Q_{t+j})$ carries Q_t as an argument for $j \leq s$ and drops it afterward. Each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \mu_{t+1}\,\Phi_1 + \frac{1}{R}\frac{\partial V_{t+2}}{\partial Q_t}, \qquad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t},$$

and iterating until Q_t drops out ($\Phi_j = 0$ for $j > s$ by construction),

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^{s} \frac{1}{R^{j-1}}\,\mu_{t+j}\,\Phi_j.$$

Substituting into (14), with $\Phi_0 = S_q$,

$$\nu_t = \mu_t\,\Phi_0 + \frac{1}{R}\sum_{j=1}^{s}\frac{1}{R^{j-1}}\,\mu_{t+j}\,\Phi_j = \sum_{j=0}^{s}\frac{1}{R^j}\,\mu_{t+j}\,\Phi_j.$$

Substituting the damage block above for each μ_{t+j} and writing the date-$(t+j)$ population integrals as expectations over future skill histories,

$$\mathrm{SCC}_t \equiv \nu_t = -\sum_{j=0}^{s} \frac{1}{R^j}\,\mathbb{E}_t \int \left[1 + \left(\frac{\tau^y_{t+j}}{1-\tau^y_{t+j}}\right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1+\epsilon_{t+j} - \gamma_{c,t+j}}\right] \frac{U_{S,t+j}}{U_{c,t+j}}\, \Phi_j\, f_{t+j}\, d\theta$$

Here, s is the number of periods emissions affect the pollution stock.

C Commodity wedge constraint derivations

This appendix derives the results of Sections 6 and 7.3 in one pass. The commodity wedge is held to a schedule $\bar{p}_t(\theta) \equiv 1+\bar{\tau}^q_t(\theta)$: Section 6 reads every condition at a flat schedule, $\bar{p}_t(\theta) = 1+\bar{\tau}_t$, whose level is chosen optimally, and Section 7.3 at an arbitrary exogenous schedule. The shadow emission price ν_t of (14) is carried throughout; the no-externality economy of Section 6's first read sets $\nu_t = 0$. Multiplier conventions follow Appendix A.1, and $\eta^q(\theta) f_t(\theta|\theta_-)$ is the multiplier on the wedge constraint. Two shorthands recur: the ad-valorem deviation and its virtual counterpart,

$$\varpi_t(\theta) \equiv \frac{\bar{\tau}^q_t(\theta) - \nu_t}{1+\bar{\tau}^q_t(\theta)}, \qquad\qquad \varpi^*_t(\theta^t) \equiv (\gamma_{c,t} - \gamma_{q,t})\, \frac{\psi U_{c,t}}{\theta f_t},$$

which at $\nu_t = 0$ and a flat schedule are the ϖ_t and ϖ^*_t of Section 6.

C.1 The problem and first-order conditions

The Hamiltonian is that of Appendix B.1 plus the constraint term $\eta^q\,[U_q/U_c - \bar{p}]\, f$. Write $\Gamma^q \equiv U_q/U_c - \bar{p}$ and Γ^q_z for its partial derivatives; each follows from the quotient rule,

$$\Gamma^q_z = \frac{U_{qz}}{U_c} - \frac{U_q}{U_c}\frac{U_{cz}}{U_c} = \frac{U_{qz}}{U_c} - \bar{p}\,\frac{U_{cz}}{U_c},$$

evaluated at the constrained allocation, where $U_q/U_c = \bar{p}$. For the numéraire, with $U_{qc}/U_c = -\sigma_{cq}/q$ and $U_{cc}/U_c = -\sigma/c$,

$$\Gamma^q_c = -\frac{\sigma_{cq}}{q} + \bar{p}\,\frac{\sigma}{c} = \bar{p}\left[\frac{\sigma}{c} - \frac{\sigma_{cq}}{\bar{p}\, q}\right] = \bar{p}\,\mathcal{I}_t, \tag{15}$$

the income-effect gap of Section 6 evaluated at $1+\tau^q = \bar{p}$. For the dirty good, define $\sigma_{q,t} \equiv -U_{qq}\, q_t/U_q$, the own-curvature of the dirty good's marginal utility, so that $U_{qq}/U_c = (U_{qq}q/U_q)(U_q/U_c)/q = -\bar{p}\,\sigma_q/q$ and

$$\Gamma^q_q = -\bar{p}\,\frac{\sigma_q}{q} + \bar{p}\,\frac{\sigma_{cq}}{q} = -\bar{p}\,\frac{\sigma_q - \sigma_{cq}}{q} \equiv -\bar{p}\,\mathcal{J}_t, \tag{16}$$

with $\mathcal{J}_t \equiv (\sigma_{q,t} - \sigma_{cq,t})/q_t$ the mirror image of $\mathcal{I}_t$. For labor, with $U_{ql}/U_c = (U_{ql}l/U_q)(U_q/U_c)/l = \bar{p}\,\gamma_q/l$ and $U_{cl}/U_c = \gamma_c/l$,

$$\Gamma^q_l = \bar{p}\,\frac{\gamma_q}{l} - \bar{p}\,\frac{\gamma_c}{l} = -\bar{p}\,\frac{\gamma_c - \gamma_q}{l}. \tag{17}$$

For the stock, with $\xi_c \equiv U_{Sc}\,c/U_S$ and $\xi_q \equiv U_{Sq}\,q/U_S$ the complementarities of the pollutant with each good, $U_{qS}/U_c = \xi_q(U_S/U_c)/q$ and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, so

$$\Gamma^q_S = \frac{U_S}{U_c}\left[\frac{\xi_q}{q} - \bar{p}\,\frac{\xi_c}{c}\right]. \tag{18}$$

The first-order conditions are those of Appendix B.1, each augmented by $\eta^q f\,\Gamma^q_z$:

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: \; f - \psi U_{cl}\frac{l}{\theta} + \eta^q f\,\Gamma^q_c = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: \; f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \eta^q f\,\Gamma^q_q = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: \; -\theta f - \phi U_l + \eta^q f\,\Gamma^q_l = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial S} &: \; \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta^q f\,\Gamma^q_S\right) d\theta = \mu,
\end{aligned}$$

with the conditions for u, w, and w_2 unchanged, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the promise-keeping chain carry over. In the no-externality economy $\nu_t = 0$ and the stock condition is absent. When the level of a flat schedule is itself chosen, one more derivative is needed; it is taken in Appendix C.9.

C.2 Implementation and the first-order approach

This subsection proves Lemma 6.1. Fix a common price $\bar{p} = 1 + \bar{\tau}$ and consider the agent's split of expenditure x,

$$\tilde{U}(x, l; \bar{\tau}) = \max_{c,q \geq 0} U(c, q, l) \quad \text{s.t.} \quad c + \bar{p}\,q = x.$$

The first-order condition of the split is the tangency $U_q = \bar{p}\, U_c$, and the second derivative of U along the budget line (substituting $dc = -\bar{p}\, dq$) is $U_{qq} - 2\bar{p}U_{cq} + \bar{p}^2 U_{cc} = \mathcal{D} < 0$, so the interior split exists and is unique. A bundle (c, q) therefore satisfies the wedge constraint $U_q/U_c = \bar{p}$ if and only if it is the split of $x = c + \bar{p}q$, which is the implementation claim.

For the envelope identities, differentiate $\tilde{U} = U(c^d, q^d, l)$. Differentiating the budget in x gives $\partial_x c^d + \bar{p}\, \partial_x q^d = 1$, so, using the tangency,

$$\tilde{U}_x = U_c\, \partial_x c^d + U_q\, \partial_x q^d = U_c \left(\partial_x c^d + \bar{p}\, \partial_x q^d\right) = U_c.$$

Differentiating the budget in l at fixed x gives $\partial_l c^d + \bar{p}\, \partial_l q^d = 0$, so

$$\tilde{U}_l = U_l + U_c \left(\partial_l c^d + \bar{p}\, \partial_l q^d\right) = U_l.$$

For the incentive-compatibility claim: a deviator who reports $\hat{\theta}$ receives the bundle assigned to $\hat{\theta}$, equivalently the pair $(x(\hat{\theta}), y(\hat{\theta}))$, works $l = y(\hat{\theta})/\theta$, and splits $x(\hat{\theta})$ at the same common price, so the deviation space is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{\tau})$. The type enters $\tilde{U}$ only through $l = y/\theta$, so the envelope condition is $\dot{u} = -\tilde{U}_l\, l/\theta + \beta w_2 = -U_l\, l/\theta + \beta w_2$, and the first-order approach applies as in the one-good economy of Appendix A.1.

C.3 Demand identities

Differentiating the tangency $U_q = \bar{p}U_c$ together with the budget gives the demand derivatives. *In x at fixed $l, \bar{p}$:* the budget gives $\partial_x c^d = 1 - \bar{p}\, \partial_x q^d$; substituting into $U_{qc}\partial_x c^d + U_{qq}\partial_x q^d = \bar{p}(U_{cc}\partial_x c^d + U_{cq}\partial_x q^d)$ and collecting,

$$\mathcal{D}\, \partial_x q^d = -\left(U_{cq} - \bar{p}U_{cc}\right) \qquad \Longrightarrow \qquad \partial_x q^d = \frac{U_{cq} - \bar{p}U_{cc}}{|\mathcal{D}|} = \frac{\bar{p}\, U_c\, \mathcal{I}_t}{|\mathcal{D}|}, \tag{19}$$

where the last step uses $U_{cq} - \bar{p}U_{cc} = U_c\, (\sigma\bar{p}/c - \sigma_{cq}/q) = \bar{p}\, U_c\, \mathcal{I}_t$ from the definitions of the statistics. Since $\partial_x q^d$ is the income expansion of dirty-good demand, $\mathcal{I}_t > 0$ exactly when the dirty good is normal; the budget then gives $\partial_x c^d = 1 - m_q$ with $m_q \equiv \bar{p}\, \partial_x q^d$, so $\mathcal{J}_t$, which satisfies $U_q \mathcal{J}_t = \bar{p}U_{cq} - U_{qq}$ and hence $\partial_x c^d = U_q \mathcal{J}_t/|\mathcal{D}|$, is positive exactly when the numéraire is normal. *In l at fixed $x, \bar{p}$:* the budget gives $\partial_l c^d = -\bar{p}\, \partial_l q^d$, and the differentiated tangency yields $\mathcal{D}\, \partial_l q^d = \bar{p}U_{cl} - U_{ql}$; with $U_{cl} = \gamma_c U_c/l$ and $U_{ql} = \gamma_q \bar{p} U_c/l$,

$$\partial_l q^d = \frac{\bar{p}\, U_c\, (\gamma_c - \gamma_q)}{l\, \mathcal{D}} = \frac{\bar{p}\, U_c\, (\gamma_q - \gamma_c)}{l\, |\mathcal{D}|}. \tag{20}$$

Compensated own-price response: along an indifference curve $dc = -\bar{p}\,dq$, and differentiating the tangency in $\bar{p}$ at constant utility gives $\mathcal{D}\,dq = U_c\,d\bar{p}$, hence

$$s_{qq} \equiv \left.\frac{\partial q^d}{\partial \bar{p}}\right|_u = \frac{U_c}{\mathcal{D}} = -\frac{U_c}{|\mathcal{D}|} < 0, \qquad \frac{\partial q^d}{\partial \bar{p}} = s_{qq} - q^d\,\partial_x q^d \quad \text{(Slutsky)}.$$

Three identities tie the constraint statistics to these demand objects. First, combining (20) with s_{qq},

$$l\,\partial_l q^d = -\bar{p}\,|s_{qq}|(\gamma_c - \gamma_q)\,. \tag{21}$$

Second, adding the two constraint curvatures, $\bar{p}\,\mathcal{I}_t = (U_{cq} - \bar{p}U_{cc})/U_c$ and $\mathcal{J}_t = (\bar{p}U_{cq} - U_{qq})/U_q = (\bar{p}U_{cq} - U_{qq})/(\bar{p}U_c)$,

$$\Delta_t^q \equiv \bar{p}\,\mathcal{I}_t + \mathcal{J}_t = \frac{\bar{p}U_{cq} - \bar{p}^2U_{cc} + \bar{p}U_{cq} - U_{qq}}{\bar{p}\,U_c} = \frac{-\mathcal{D}}{\bar{p}\,U_c} = \frac{1}{\bar{p}\,|s_{qq}|} > 0. \tag{22}$$

Third, combining (19) and (22), the marginal budget share is the numéraire-side curvature share,

$$m_q = \bar{p}\,\partial_x q^d = \frac{\bar{p}^2\,U_c\,\mathcal{I}_t}{|\mathcal{D}|} = \frac{\bar{p}\,\mathcal{I}_t}{\Delta_t^q}. \tag{23}$$

C.4 The multiplier

Dividing the first-order condition for c by $U_c f$, using $U_{cl}l = \gamma_c U_c$ and (15),

$$\frac{U_c\,\phi}{f} = 1 - \gamma_c\frac{\psi U_c}{\theta f} + \eta^q\,\bar{p}\,\mathcal{I}_t. \tag{24}$$

Dividing the first-order condition for q by $U_q f$, multiplying by U_c, and substituting $U_c/U_q = 1/\bar{p}$ and (16),

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{\bar{p}} - \gamma_q\frac{\psi U_c}{\theta f} + \eta^q\,\frac{\Gamma_q^q}{\bar{p}} = \frac{1+\nu_t}{\bar{p}} - \gamma_q\frac{\psi U_c}{\theta f} - \eta^q\,\mathcal{J}_t. \tag{25}$$

Equating (24) and (25) and collecting,

$$1-\frac{1+\nu_t}{\bar{p}}=(\gamma_c-\gamma_q)\frac{\psi U_c}{\theta f}-\eta^q\left(\bar{p}\,\mathcal{I}_t+\mathcal{J}_t\right),$$

and since $1-(1+\nu_t)/\bar{p}=(\bar{\tau}^q_t-\nu_t)/(1+\bar{\tau}^q_t)=\varpi_t$, with Δ^q_t from (22),

$$\eta^q_t\,\Delta^q_t=\varpi^*_t-\varpi_t \qquad\Longleftrightarrow\qquad \eta^q_t=\bar{p}_t\,|s_{qq,t}|(\varpi^*_t-\varpi_t)\,. \tag{26}$$

At $\nu_t=0$ and a flat schedule this is Lemma 6.2; with the externality it is Lemma 7.3 with $\mathrm{SCC}_t\equiv\nu_t$. At $\eta^q_t=0$, (26) is the primitive corrective-wedge condition of Appendix B.1, so the multiplier vanishes exactly where the frozen deviation equals the virtual one.

C.5 Labor wedge

From the first-order condition for l, using $U_{ll}l+U_l=U_l(1+\epsilon)$,

$$\phi=-\frac{\theta f}{U_l}-\frac{\psi}{\theta}(1+\epsilon)+\frac{\eta^q f\,\Gamma^q_l}{U_l}.$$

With $U_l=-(1-\tau^y_t)\theta U_c$, where τ^y_t is the endogenous labor wedge, and (17),

$$\frac{\Gamma^q_l}{U_l}=\frac{-\bar{p}(\gamma_c-\gamma_q)/l}{-(1-\tau^y_t)\theta U_c}=\frac{\bar{p}\,(\gamma_c-\gamma_q)}{(1-\tau^y_t)\,y\,U_c},$$

so multiplying by U_c/f,

$$\frac{U_c\,\phi}{f}=\frac{1}{1-\tau^y_t}-(1+\epsilon)\frac{\psi U_c}{\theta f}+\eta^q\,\frac{\bar{p}\,(\gamma_c-\gamma_q)}{(1-\tau^y_t)\,y}. \tag{27}$$

Equating (24) and (27) and collecting, with $1/(1-\tau^y_t)-1=\tau^y_t/(1-\tau^y_t)$ and $A_t=1+\epsilon-\gamma_c$,

$$\frac{\tau^y_t}{1-\tau^y_t}=A_t\frac{\psi U_c}{\theta f}+\eta^q_t\,\bar{p}_t\left[\mathcal{I}_t-\frac{\gamma_{c,t}-\gamma_{q,t}}{(1-\tau^y_t)\,y_t}\right], \tag{28}$$

the primitive labor wedge of Results 6.1 and 7.6. The bracket is the response of the implied commodity wedge $\ln(U_q/U_c)$ to a utility-compensated unit of earnings, per net-of-tax dollar: along $(dc,dl)=(1-\tau^y_t,\,1/\theta)$, $d\ln(U_q/U_c)=(1-\tau^y_t)\,\mathcal{I}_t+(\gamma_q-\gamma_c)/y$, and dividing by $1-\tau^y_t$

gives the bracket. Under weak separability the bracket is $\mathcal{I}_t$, and using (26) with $\varpi^* = 0$ and (23),

$$\eta_t^q \, \bar{p}_t \, \mathcal{I}_t = -\frac{\bar{p}_t \, \mathcal{I}_t}{\Delta_t^q} \, \varpi_t = -m_{q,t} \, \varpi_t = m_{q,t} \, \frac{\nu_t - \bar{\tau}_t^q}{1 + \bar{\tau}_t^q},$$

which is Corollary 7.3.

C.6 Savings wedge

Define $\chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \bar{p}_t \mathcal{I}_t$, so that (24) reads $\phi/f = (1 - \chi_t^q)/U_{c,t}$. The promise-keeping chain of Appendix A.2 is untouched: the first-order condition for w and the envelope condition give $\lambda_{1,t+1} = \beta R(1 - \chi_t^q)/U_{c,t}$, integrating the costate equation over $(0, \infty)$ gives $\lambda_{1,t} = \int \phi \, d\theta = \mathbb{E}_{t-1}[(1 - \chi_t^q)/U_{c,t}]$, and equating the two one period apart,

$$\frac{1 - \chi_t^q}{U_{c,t}} = \frac{1}{\beta R} \, \mathbb{E}_t \left[\frac{1 - \chi_{t+1}^q}{U_{c,t+1}} \right].$$

Substituting the multiplier (26) and the curvature-share identity (23) assembles the adjustment into composite form,

$$\chi_t^q = \gamma_c \frac{\psi U_c}{\theta f} - m_q \left(\varpi_t^* - \varpi_t \right) = \left[\gamma_c - m_q (\gamma_c - \gamma_q) \right] \frac{\psi U_c}{\theta f} + m_q \, \varpi_t = \tilde{\gamma}_t \, \frac{\psi U_{c,t}}{\theta f_t} + m_{q,t} \, \varpi_t,$$

with $\tilde{\gamma} \equiv (1 - m_q)\gamma_c + m_q \gamma_q$ as in Lemma 6.3; at $\nu_t = 0$ this is Result 6.3's $\chi^F = \tilde{\gamma} \, \psi U_c/(\theta f) + \varpi m_q$, and $m_q \varpi = (\bar{\tau}^q - \nu) \, \partial_x q^d$ is the commodity revenue, net of marginal damages, returned by a marginal dollar of expenditure. The exact wedge, its second-order approximation, and the lifetime formula follow the steps of Appendix A.2 with $\chi_t \mapsto \chi_t^q$; in particular $\tau_t^s \approx \mathrm{Var}_t(\ln U_{c,t+1}) + (\chi_t^q - \mathbb{E}_t \, \chi_{t+1}^q) + \mathrm{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1})$.

C.7 Composite statistics and the comprehensive wedge

Substituting the multiplier (26) into the labor condition (28) yields the comprehensive form of Result 6.1. Using $\bar{p}/\Delta^q = \bar{p}^2 |s_{qq}|$ from (22), the correction term expands into four pieces,

$$\eta^q \bar{p} \left[\mathcal{I} - \frac{\gamma_c - \gamma_q}{(1 - \tau^y) y} \right] = \left[m_q - \frac{\bar{p}^2 |s_{qq}| (\gamma_c - \gamma_q)}{(1 - \tau^y) y} \right] \left(\varpi^* - \varpi \right),$$

and, expanding with $\varpi^* = (\gamma_c - \gamma_q) \, \psi U_c/(\theta f)$,

$$\begin{aligned} m_q\,\varpi^* &= m_q(\gamma_c-\gamma_q)\frac{\psi U_c}{\theta f} = (\gamma_c-\tilde\gamma)\frac{\psi U_c}{\theta f},\\ -\frac{\bar p^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\,\varpi^* &= -\frac{\bar p^2|s_{qq}|(\gamma_c-\gamma_q)^2}{(1-\tau^y)y}\,\frac{\psi U_c}{\theta f} = -\left(\epsilon-\tilde\epsilon\right)\frac{\psi U_c}{\theta f},\\ -m_q\,\varpi &= -\left(\bar\tau^q-\nu\right)\partial_x q^d,\\ +\frac{\bar p^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\,\varpi &= -\frac{\bar\tau^q-\nu}{1-\tau^y}\cdot\frac{\partial_l q^d}{\theta}, \end{aligned}$$

where the second line uses $l|U_l|=(1-\tau^y)\theta l\,U_c=(1-\tau^y)y\,U_c$ to match Lemma 6.3's Le Chatelier correction, $\epsilon-\tilde\epsilon=\bar p^2U_c^2(\gamma_c-\gamma_q)^2/(l|U_l||\mathcal{D}|)=\bar p^2|s_{qq}|(\gamma_c-\gamma_q)^2/((1-\tau^y)y)$, and the fourth line uses $\partial_l q^d/\theta=-\bar p|s_{qq}|(\gamma_c-\gamma_q)/y$ from (21) and $\varpi\bar p=\bar\tau^q-\nu$. Moving the two fiscal pieces to the left-hand side of (28) and collecting the $\psi U_c/(\theta f)$ terms on the right,

$$\frac{\tau_t^y}{1-\tau_t^y}+\frac{\bar\tau_t^q-\nu_t}{1-\tau_t^y}\left[(1-\tau_t^y)\,\partial_x q^d+\frac{\partial_l q^d}{\theta}\right]=\left[A_t+(\gamma_c-\tilde\gamma)-(\epsilon-\tilde\epsilon)\right]\frac{\psi U_c}{\theta f}=\tilde A_t\,\frac{\psi U_c}{\theta f},$$

which is the comprehensive form, with $\bar\tau^q-\nu$ the screening tax and the bracket the compensated demand response $dq^d/dy|_u$: along a perturbation dy with $dx=(1-\tau^y)dy$, utility is unchanged and $dq^d=[(1-\tau^y)\partial_x q^d+\partial_l q^d/\theta]dy$.

The composite statistics are also the curvatures of the indirect utility of Lemma 6.1, which proves the second claim of Lemma 6.3. Since $\tilde U_x=U_c(c^d,q^d,l)$ and $\partial_l c^d=-\bar p\,\partial_l q^d$,

$$\tilde U_{xl}=U_{cl}+(U_{cq}-\bar p\,U_{cc})\,\partial_l q^d,$$

and dividing by $\tilde U_x=U_c$, multiplying by l, and substituting (19) and (20),

$$\frac{\tilde U_{xl}\,l}{\tilde U_x}=\gamma_c+\frac{l\,(U_{cq}-\bar pU_{cc})}{U_c}\cdot\frac{\bar pU_c(\gamma_q-\gamma_c)}{l\,|\mathcal{D}|}=\gamma_c+m_q\,(\gamma_q-\gamma_c)=\tilde\gamma.$$

Similarly $\tilde U_l=U_l(c^d,q^d,l)$ gives, using $U_{lq}-\bar pU_{lc}=(\bar pU_c/l)(\gamma_q-\gamma_c)$,

$$\tilde U_{ll}=U_{ll}+(U_{lq}-\bar p\,U_{lc})\,\partial_l q^d=U_{ll}+\frac{\bar p^2U_c^2(\gamma_c-\gamma_q)^2}{l^2\,|\mathcal{D}|},$$

and dividing by $\tilde U_l=U_l<0$ flips the sign of the positive correction,

$$\frac{\tilde U_{ll}\, l}{\tilde U_l} = \epsilon - \frac{\bar p^2\, U_c^2\, (\gamma_c - \gamma_q)^2}{l\, |U_l|\, |\mathcal{D}|} = \tilde\epsilon \le \epsilon.$$

Under weak separability $\gamma_c = \gamma_q$, both corrections vanish, and the restricted problem is that of Appendix A.1 exactly.

C.8 The costate and the comprehensive-wedge decomposition

Substituting $\phi = f/U_c - \psi\gamma_c/\theta + \eta^q \bar p \mathcal{I}_t\, f/U_c$ from (24) into the costate equation, and eliminating the multiplier with $\eta^q \bar p \mathcal{I}_t = m_q(\varpi^* - \varpi)$ and $m_q \varpi^*\, f/U_c = m_q(\gamma_c - \gamma_q)\, \psi/\theta$,

$$\dot\psi - \frac{\tilde\gamma}{\theta}\, \psi = \lambda_1 \bar\alpha_t f - \lambda_2 f_2 - [1 - m_q\, \varpi]\, \frac{f}{U_c},$$

the ODE of Appendix A.1 with the kernel $\tilde\gamma = \gamma_c - m_q(\gamma_c - \gamma_q)$ and the source weight $1 - m_q \varpi = 1 - (\bar\tau^q - \nu)\, \partial_x q^d$: the resource value of taking a dollar from a type nets out the commodity revenue, in excess of marginal damages, that the dollar's spending would have returned. Solving with the integrating factor $\exp[-\int \tilde\gamma\, d\tilde x/\tilde x]$ and $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \tilde\gamma(\tilde x) \frac{d\tilde x}{\tilde x}\right] \left([1 - m_q(x)\, \varpi(x)]\, \frac{f(x)}{U_c(x)} - \lambda_1 \bar\alpha_t(x) f(x) + \lambda_2 f_2(x) \right) dx,$$

with $\psi(0) = 0$ pinning $\lambda_{1,t}$ as in (9). To carry $U_c(\theta)$ into the integrand, multiply the identity (10) by $\exp[\int_\theta^x (\gamma_c - \tilde\gamma)\, d\tilde x/\tilde x]$, with $\gamma_c - \tilde\gamma = m_q(\gamma_c - \gamma_q)$:

$$\frac{U_c(\theta)}{U_c(x)} \exp\left[-\int_\theta^x \tilde\gamma(\tilde x) \frac{d\tilde x}{\tilde x}\right] = \exp\left[\int_\theta^x \left(\sigma \frac{\dot c}{c} + \sigma_{cq} \frac{\dot q}{q} - \gamma_c \frac{\dot y}{y} + m_q\, (\gamma_c - \gamma_q)\, \frac{1}{\tilde x} \right) d\tilde x \right] \equiv \tilde K(\theta, x).$$

By Appendix C.7, $\Omega_t = \tilde A_t\, \psi U_c/(\theta f)$, so multiplying the costate solution by $\tilde A_t U_c/(\theta f)$ and splitting the integral as in Appendix A.1,

$$\Omega_t = \tilde A_t(\theta)\, \tilde B_t(\theta)\, \tilde C_t(\theta) + \beta R\, \Omega_{t-1}\, \tilde D_t(\theta),$$

$$\tilde{B}_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$\tilde{C}_t(\theta) = \int_\theta^\infty \tilde{K}(\theta, x)\,(1 - m_q(x)\,\varpi(x) - \lambda_{1,t}\bar{\alpha}_t(x) U_{c,t}(x))\,\frac{f_t(x)dx}{1 - F_t(\theta)}$$

$$\tilde{D}_t(\theta) = \frac{\tilde{A}_t(\theta)}{\tilde{A}_{t-1}}\,\frac{U_{c,t}(\theta)}{U_{c,t-1}}\,\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \tilde{\gamma}_t(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)},$$

where the persistence block uses $\lambda_{2,t} = \beta R\,\psi_{t-1}/f_{t-1}$ and, from $\Omega_{t-1} = \tilde{A}_{t-1}\,\psi_{t-1}U_{c,t-1}/(\theta_{t-1}f_{t-1})$,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\theta_{t-1}\,\Omega_{t-1}}{\tilde{A}_{t-1}\,U_{c,t-1}},$$

so the recursing object is the comprehensive wedge, which is Result 6.4. The revenue term $-m_q\varpi$ inside the $\tilde{C}_t$ weight is the only trace of the second good in the redistributive valuation.

C.9 The optimal uniform price

Let the schedule be flat, $\bar{\tau}_t^q(\theta) = \bar{\tau}_t$, with the level chosen by the planner. The level enters the date-0 Lagrangian only through the date-t wedge constraints, each contributing $\eta^q f\,\partial[U_q/U_c - (1 + \bar{\tau}_t)]/\partial\bar{\tau}_t = -\eta^q f$; the R^{-t} discounting and the transition densities convert the sum over date-t nodes into the unconditional population expectation, as in the aggregation of Appendix B.1. By the envelope theorem, then,

$$\frac{dW_0}{d\bar{\tau}_t} = -\frac{1}{R^t}\,\mathbb{E}_0[\eta_t^q]\,,$$

so the optimal level sets $\mathbb{E}_0[\eta_t^q] = 0$. In the reduction bookkeeping the same condition emerges from Roy's identity offsetting the incentive value of compensation; the constraint formulation nets the two automatically, and only the substitution distortion against the screening benefit survives. Substituting (26) with $\bar{p}_t$ and ϖ_t common across the date-t cross-section,

$$\mathbb{E}_0[\eta_t^q] = \bar{p}_t\,\{\mathbb{E}_0[|s_{qq,t}|\,\varpi_t^*] - \varpi_t\,\mathbb{E}_0[|s_{qq,t}|]\} = 0 \qquad \Longrightarrow \qquad \varpi_t = \frac{\mathbb{E}_0\left[|s_{qq,t}|\,(\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]},$$

which at $\nu_t = 0$ is the ad-valorem form of Result 6.2 and, with the externality, the split of Result 6.6, $1 + \tau_t^q = 1 + \mathrm{SCC}_t + \bar{\tau}_t^{\text{screen}}$ with $\bar{\tau}_t^{\text{screen}}/(1 + \tau_t^q) = \varpi_t$. The per-unit form follows from (21): multiplying the condition by $\bar{p}_t\,\mathbb{E}_0[|s_{qq}|]$ and substituting $\bar{p}|s_{qq}|(\gamma_c-\gamma_q)\,\psi U_c/(\theta f) = -\,(\psi U_c/(\theta f))\,l\,\partial_l q^d$ and $\bar{p}\varpi_t = \bar{\tau}_t - \nu_t$,

$$(\bar{\tau}_t-\nu_t)\,\mathbb{E}_0[|s_{qq,t}|] = -\,\mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\,l_t\,\partial_l q_t^d\right] \qquad \Longleftrightarrow \qquad (\bar{\tau}_t-\nu_t)\,\mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\,l_t\,\partial_l q_t^d\right],$$

the first display of Result 6.2 at $\nu_t = 0$. At node-specific prices the same condition without the integral gives $\varpi(\theta^t) = \varpi^*(\theta^t)$, the unconstrained wedge of Appendix A.3; the uniform tax is its $|s_{qq}|$-weighted average. Under weak separability every node wedge is zero, so $\bar{\tau}_t = \nu_t$ (Corollaries 6.1 and 7.3); in retirement $\psi \equiv 0$, so $\psi U_c/(\theta f) \equiv 0$ and $\bar{\tau}_t = \nu_t$ (Corollary 6.2).

C.10 The cost of uniformity

By the envelope theorem applied node by node, the date-t resource value of moving a single node's price is $-\eta^q(\theta)f(\theta)$ per unit of $\bar{\tau}^q(\theta)$, and by (26),

$$-\eta^q\,f = \bar{p}\,|s_{qq}|(\varpi-\varpi^*)\,f,$$

which vanishes at $\varpi = \varpi^*$, reproducing the node-specific optimum. The marginal resource cost of moving the node's ad-valorem wedge is $\bar{p}^2$ times this, since $\varpi = 1-(1+\nu)/\bar{p}$ gives $d\bar{\tau}^q = \bar{p}^2\,d\varpi/(1+\nu)$, which at $\nu = 0$ is $d\bar{\tau} = \bar{p}^2 d\varpi$. The loss from setting ϖ rather than ϖ^* at a node is then, to second order (the envelope theorem killing the first-order effects of the induced adjustments in the allocation and the multipliers, and $\bar{p}$, s_{qq}, ϖ^* held at their values at the optimum),

$$\int_{\varpi^*}^{\varpi} \bar{p}^3\,|s_{qq}|\,(v-\varpi^*)\,dv = \frac{1}{2}\,\bar{p}^3\,|s_{qq}|\,(\varpi-\varpi^*)^2\,.$$

Aggregating over the date-t cross-section, with $\bar{p}$ common because the price is uniform, and minimizing the resulting quadratic over the common ϖ yields the weighted mean $\varpi = \mathbb{E}_t^{|s|}[\varpi_t^*]$ of Appendix C.9 and, at that minimizer,

$$\mathrm{Loss}_t \approx \frac{1}{2}\,(1+\bar{\tau}_t)^3\,\mathbb{E}_0[|s_{qq,t}|]\,\mathrm{Var}_t^{|s|}(\varpi_t^*)\,,$$

where $\mathbb{E}_t^{|s|}$ and $\mathrm{Var}_t^{|s|}$ are the $|s_{qq,t}|$-weighted population mean and variance over date-t histories, which is Result 6.5.

C.11 The externality and the social cost of carbon

The conditions above already carry the shadow emission price: every deviation is measured net of ν_t, so all of Section 6's results hold with $\bar{\tau}_t = \tau_t^q - \nu_t$ the screening tax and every statistic evaluated at the externality allocation, and the optimal flat level of Appendix C.9 delivers the additive split of Result 6.6. What remains is the stock multiplier. Substitute the three pieces of the first-order condition for S,

$$-\psi U_{Sl}\frac{l}{\theta} = -\frac{\psi U_c}{\theta f}\,\xi\,\frac{U_S}{U_c}\,f, \qquad -\phi U_S = -\,(1-\chi_t^q)\,\frac{U_S}{U_c}\,f, \qquad \eta^q f\,\Gamma_S^q = \eta^q\,\frac{U_S}{U_c}\left[\frac{\xi_q}{q} - \bar{p}\,\frac{\xi_c}{c}\right] f,$$

the second from (24) and the third from (18), to get

$$\mu_t = -\int_0^\infty \left[\frac{\psi U_c}{\theta f}\,\xi + 1 - \chi_t^q - \eta^q\left(\frac{\xi_q}{q} - \bar{p}\,\frac{\xi_c}{c}\right)\right]\frac{U_S}{U_c}\,f\,d\theta.$$

Expanding $1-\chi_t^q = 1 - \gamma_c\frac{\psi U_c}{\theta f} + \eta^q\bar{p}\,\mathcal{I}_t$ with $\bar{p}\,\mathcal{I}_t = \bar{p}\,\sigma/c - \sigma_{cq}/q$ and collecting,

$$\mu_t = -\int_0^\infty \left[1 + (\xi-\gamma_c)\,\frac{\psi U_c}{\theta f} + \eta^q\left(\bar{p}\,\frac{\sigma+\xi_c}{c} - \frac{\sigma_{cq}+\xi_q}{q}\right)\right]\frac{U_S}{U_c}\,f\,d\theta,$$

and the envelope iteration $\nu_t = \sum_{j=0}^{s} R^{-j}\mu_{t+j}\Phi_j$ of Appendix B.1 gives the damage weight of Result 7.6. The new piece is the response of the damage valuation to a budget-neutral clean-for-dirty swap: along $(dc, dq) = (\bar{p}, -1)$, which holds expenditure at consumer prices fixed, $d\ln(U_S/U_c) = \bar{p}(\sigma+\xi_c)/c - (\sigma_{cq}+\xi_q)/q$, using $\partial_c \ln(U_S/U_c) = (\sigma+\xi_c)/c$ and $\partial_q \ln(U_S/U_c) = (\sigma_{cq}+\xi_q)/q$. At $\eta^q \equiv 0$ the weight collapses to the fiscal factor of Appendix B.1; at the optimal flat level the multiplier averages to zero but is not zero node by node, so the weight carries the term even in Section 6, which is the sense in which Result 6.6 says the fiscal factor gains a term proportional to η^q. The corrective component of the price is nonetheless uniform across types and histories, because damages depend only on aggregate emissions, so the emissions margin carries no information about types and cannot screen; the uniformity restriction binds only the Corlett-Hague screening component and never the Pigouvian one.

D Third-best derivations: the frozen labor wedge

The modified problem's Hamiltonian is

$$
\begin{aligned}
\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, S)\right] f \\
& + \psi \left[-U_l(c, q, y/\theta, S)\frac{l}{\theta} + \beta w_2\right] \\
& - \lambda_1 u(\theta)\bar{\alpha}_t f_t \\
& + \lambda_2 u(\theta) f_2 \\
& + \phi \left[u - U(c, q, y/\theta, S) - \beta w\right] \\
& - \mu \left[S - S(\ldots, Q)\right] \\
& + \eta \left[-\frac{U_l(c, q, y/\theta, S)}{\theta\, U_c(c, q, y/\theta, S)} - (1 - \bar{\tau}_t^y(\theta))\right] f.
\end{aligned}
$$

Write $\Gamma \equiv -U_l/(\theta U_c) - (1 - \bar{\tau}_t^y(\theta))$ for the constraint and $\Gamma_c, \Gamma_q, \Gamma_l, \Gamma_S$ for its partial derivatives. The first-order conditions are as follows:

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: \; f - \psi U_{cl}\frac{l}{\theta} + \eta f\, \Gamma_c = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: \; f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \eta f\, \Gamma_q = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: \; -\theta f - \phi U_l + \eta f\, \Gamma_l = \frac{1}{\theta}\psi \left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \; \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \; \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta f\, \Gamma_S\right) d\theta = \mu,
\end{aligned}
$$

with $\nu_t \equiv \mu_t S_q + \frac{1}{R}\partial V_{t+1}/\partial Q$. Each derivative of Γ can be expressed as follows:

$$
\begin{aligned}
\Gamma_c &= (1-\bar{\tau}_t^y)\frac{\sigma}{c} - \frac{\gamma_c}{y} \\
\Gamma_q &= (1-\bar{\tau}_t^y)\frac{\sigma_{cq}}{q} - \frac{\gamma_q(1+\tau^q)}{y} \\
\Gamma_l &= (1-\bar{\tau}_t^y)\frac{\epsilon-\gamma_c}{l} \\
\Gamma_S &= -\frac{U_S}{U_c}\left[\frac{\xi}{y} + (1-\bar{\tau}_t^y)\frac{\xi_c}{c}\right]
\end{aligned}
$$

$\xi_c \equiv U_{Sc}c/U_S$ is the elasticity of the pollutant with respect to the numéraire consumption good. From the first-order condition for c,

$$
\phi = \frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c + \eta\,\frac{\Gamma_c}{U_c}f \tag{29}
$$

From the first-order condition for l,

$$
\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \eta\frac{\Gamma_l}{U_l}f
$$

Hence,

$$
\frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c + \eta\,\frac{\Gamma_c}{U_c}f = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \eta\frac{\Gamma_l}{U_l}f
$$

After collecting terms,

$$
\left(\frac{U_l}{\theta U_c} + 1\right)\theta f = -\frac{\psi U_l}{\theta}(1+\epsilon-\gamma_c) + \eta U_l\left[\frac{\Gamma_l}{U_l} - \frac{\Gamma_c}{U_c}\right]f
$$

Because $\bar{\tau}_t^y = 1 + \frac{U_l}{\theta U_c}$,

$$
\bar{\tau}_t^y\theta f = -\frac{\psi U_l}{\theta}(1+\epsilon-\gamma_c) + \eta U_l\left[\frac{\Gamma_l}{U_l} - \frac{\Gamma_c}{U_c}\right]f
$$

Dividing by θf, multiplying the RHS by $\frac{U_c}{U_c}$, and using the wedge definition,

$$\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} = \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) - \eta\left[\Gamma_l \frac{U_c}{U_l} - \Gamma_c\right]$$

Substituting Γ_l by its definition,

$$\begin{aligned}
\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} &= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) - \eta\left[(1-\bar{\tau}_t^y)\frac{\epsilon-\gamma_c}{l}\frac{U_c}{U_l} - (1-\bar{\tau}_t^y)\frac{\sigma}{c} + \frac{\gamma_c}{y}\right] \\
&= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) - \eta\left[(1-\bar{\tau}_t^y)\frac{\epsilon-\gamma_c}{y}\frac{\theta U_c}{U_l} - (1-\bar{\tau}_t^y)\frac{\sigma}{c} + \frac{\gamma_c}{y}\right] \\
&= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) + \eta\left[\frac{\epsilon-\gamma_c}{y} + (1-\bar{\tau}_t^y)\frac{\sigma}{c} - \frac{\gamma_c}{y}\right] \\
&= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) + \eta\left[(1-\bar{\tau}_t^y)\frac{\sigma}{c} + \frac{\epsilon-2\gamma_c}{y}\right]
\end{aligned}$$

D.1 Savings wedge

Define $\chi_t^\eta \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - \eta_t \Gamma_{c,t}$. Then (29) reads

$$\frac{\phi}{f} = \frac{1-\chi_t^\eta}{U_{c,t}} \tag{30}$$

The chain of Appendix A.2 is unchanged, because the constraint enters none of its steps. The first-order condition for w and the envelope condition $\partial V_{t+1}/\partial \hat{w} = \lambda_{1,t+1}$ give

$$\lambda_{1,t+1}(\theta^t) = \beta R\,\frac{\phi(\theta)}{f(\theta)} = \beta R\,\frac{1-\chi_t^\eta}{U_{c,t}}.$$

Integrating the costate equation with $\psi(0) = \psi(\infty) = 0$, $\int \bar{\alpha}_t f\, d\theta = 1$, and $\int f_2\, d\theta = 0$,

$$\lambda_{1,t} = \int_0^\infty \phi\, d\theta = \mathbb{E}_{t-1}\left[\frac{1-\chi_t^\eta}{U_{c,t}}\right].$$

Evaluating the second expression at $t+1$ and equating with the first,

$$\frac{1-\chi_t^{\eta}}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^{\eta}}{U_{c,t+1}}\right]$$

This is the adjusted generalized inverse Euler equation.

D.2 Commodity wedge

Dividing the first-order condition for q by U_q, using $U_{ql}l = \gamma_q U_q$,

$$\phi = \frac{(1+\nu_t)f}{U_q} - \frac{\psi}{\theta}\gamma_q + \eta\,\frac{\Gamma_q}{U_q}f$$

Multiplying by $\frac{U_c}{f}$, with $U_c/U_q = 1/(1+\tau^q)$,

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{1+\tau^q} - \frac{\psi U_c}{\theta f}\gamma_q + \eta\,\frac{\Gamma_q}{1+\tau^q}$$

Substituting Γ_q by its definition,

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{1+\tau^q} - \frac{\psi U_c}{\theta f}\gamma_q + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} - \frac{\gamma_q}{y}\right]$$

The same operations on the FOC for c, (29), with Γ_c substituted by its definition yields

$$\frac{U_c\,\phi}{f} = 1 - \frac{\psi U_c}{\theta f}\gamma_c + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma}{c} - \frac{\gamma_c}{y}\right]$$

Hence,

$$1 - \frac{\psi U_c}{\theta f}\gamma_c + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma}{c} - \frac{\gamma_c}{y}\right] = \frac{1+\nu_t}{1+\tau^q} - \frac{\psi U_c}{\theta f}\gamma_q + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} - \frac{\gamma_q}{y}\right]$$

Moving every term except $\frac{1+\nu_t}{1+\tau^q}$ to the left,

$$1 - \frac{1+\nu_t}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c-\gamma_q) + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} - \frac{\gamma_q}{y} - (1-\bar{\tau}_t^y)\,\frac{\sigma}{c} + \frac{\gamma_c}{y}\right]$$

Grouping the $1/y$ terms with the complementarity gap and the curvature terms with $1-\bar{\tau}_t^y$,

$$1-\frac{1+\nu_t}{1+\tau^q}=(\gamma_c-\gamma_q)\left[\frac{\psi U_c}{\theta f}+\frac{\eta}{y}\right]-\eta\,(1-\bar{\tau}_t^y)\left[\frac{\sigma}{c}-\frac{\sigma_{cq}}{(1+\tau^q)\,q}\right]$$

Because $1-\frac{1+\nu_t}{1+\tau^q}=\frac{\tau^q-\nu_t}{1+\tau^q}$ and $\mathcal{I}_t\equiv\frac{\sigma}{c}-\frac{\sigma_{cq}}{(1+\tau^q)\,q}$,

$$\frac{\tau_t^q-\nu_t}{1+\tau_t^q}=(\gamma_{c,t}-\gamma_{q,t})\left[\frac{\psi U_{c,t}}{\theta f_t}+\frac{\eta_t}{y_t}\right]-\eta_t\,(1-\bar{\tau}_t^y)\,\mathcal{I}_t$$

This is Result 7.2 with $\mathrm{SCC}_t\equiv\nu_t$ (Appendix D.4). At $\eta_t=0$ the formula reduces to the primitive condition $\frac{\tau^q-\nu_t}{1+\tau^q}=\frac{\psi U_c}{\theta f}(\gamma_c-\gamma_q)$ of Appendix B.1.

The sign of $\mathcal{I}_t$ is the normality of the dirty good. By the demand identity (19) of Appendix C.3 evaluated at the price $1+\tau_t^q$, the income expansion of dirty-good demand is $\partial_x q^d=(1+\tau^q)\,U_c\,\mathcal{I}_t/|\mathcal{D}|$, so $\mathcal{I}_t>0$ exactly when the dirty good is normal.

D.3 The costate

Substituting (29) into the costate equation $\dot{\psi}=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\phi$,

$$\dot{\psi}=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\frac{f}{U_c}+\frac{\psi}{\theta}\gamma_c-\eta\,\frac{\Gamma_c}{U_c}f$$

Moving the γ_c term to the left,

$$\dot{\psi}-\frac{\gamma_c}{\theta}\psi=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\frac{f}{U_c}-\eta\,\frac{\Gamma_c}{U_c}f$$

The multiplier is itself a function of the costate. Solving Lemma 7.1 for η,

$$\eta=\frac{1}{\Delta_t}\left[\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}-A_t\,\frac{\psi U_c}{\theta f}\right]$$

Hence,

$$\eta\,\frac{\Gamma_c}{U_c}f=\frac{\Gamma_c}{\Delta_t}\left[\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}-A_t\,\frac{\psi U_c}{\theta f}\right]\frac{f}{U_c}=\frac{\Gamma_c}{\Delta_t}\,\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}\,\frac{f}{U_c}-\frac{\Gamma_c A_t}{\Delta_t}\,\frac{\psi}{\theta}$$

Substituting into the ordinary differential equation, moving the new ψ term to the left, and defining the constraint-augmented complementarity $\gamma^\eta_{c,t} \equiv \gamma_{c,t} + A_t\,\Gamma_{c,t}/\Delta_t$,

$$\dot{\psi} - \frac{\gamma^\eta_c}{\theta}\psi = \lambda_1\bar{\alpha}_t f - \lambda_2 f_2 - \left[1 + \frac{\Gamma_c}{\Delta_t}\,\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t}\right]\frac{f}{U_c}$$

This is a linear first-order ordinary differential equation of the same form as in Appendix A.1, with γ^η_c in place of γ_c in the homogeneous part and the frozen schedule entering the source. Multiplying by the integrating factor $\exp\left[-\int^\theta \gamma^\eta_c(\tilde{x})\,d\tilde{x}/\tilde{x}\right]$ and integrating from θ to ∞ with $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \gamma^\eta_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\left[1 + \frac{\Gamma_c}{\Delta_t}\,\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t}\right]\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx$$

The boundary condition $\psi(0) = 0$ pins down $\lambda_{1,t}$ exactly as in (9), with γ^η_c in the kernel and the bracketed source in place of 1. The threat-keeping multiplier is unchanged in form, $\lambda_{2,t} = \beta R\,\psi_{t-1}/f_{t-1}$. Because Lemma 7.1 at $t-1$ gives

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\theta_{t-1}}{U_{c,t-1}}\,\frac{\psi_{t-1}\,U_{c,t-1}}{\theta_{t-1}f_{t-1}} = \frac{\theta_{t-1}}{A_{t-1}\,U_{c,t-1}}\left[\frac{\bar{\tau}^y_{t-1}}{1-\bar{\tau}^y_{t-1}} - \eta_{t-1}\Delta_{t-1}\right],$$

it reads

$$\lambda_{2,t} = \beta R\,\frac{\theta_{t-1}}{A_{t-1}\,U_{c,t-1}}\left[\frac{\bar{\tau}^y_{t-1}}{1-\bar{\tau}^y_{t-1}} - \eta_{t-1}\Delta_{t-1}\right]$$

The persistence channel loads on the frozen wedge net of the schedule's error, rather than on the optimal wedge. Through ψ, the ratio $\psi U_c/(\theta f)$ inherits the hazard and persistence structure of the ABCD decomposition, evaluated in the third best; the wedges of Section 7 then follow from it and Lemma 7.1 without further integration.

D.4 The social cost of carbon

The first-order condition for S determines the date-t stock multiplier,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta f\,\Gamma_S\right)d\theta$$

Each piece can be expressed in the marginal damage valuation U_S/U_c. Using $U_{Sl}l = \xi U_S$,

$$-\psi U_{Sl}\frac{l}{\theta} = -\frac{\psi}{\theta}\,\xi\,U_S = -\frac{\psi U_c}{\theta f}\,\xi\,\frac{U_S}{U_c}\,f$$

From (30),

$$-\phi U_S = -\left(1-\chi_t^{\eta}\right)\frac{U_S}{U_c}\,f$$

Substituting Γ_S by its definition,

$$\eta f\,\Gamma_S = -\eta\left[\frac{\xi}{y} + \left(1-\bar{\tau}_t^y\right)\frac{\xi_c}{c}\right]\frac{U_S}{U_c}\,f$$

Hence,

$$\mu_t = -\int_0^\infty \left[\frac{\psi U_c}{\theta f}\,\xi + 1 - \chi_t^{\eta} + \eta\,\frac{\xi}{y} + \eta\left(1-\bar{\tau}_t^y\right)\frac{\xi_c}{c}\right]\frac{U_S}{U_c}\,f\,d\theta$$

Expanding $1-\chi_t^{\eta} = 1-\gamma_c\frac{\psi U_c}{\theta f} + \eta\,\Gamma_c$ with Γ_c written out, the bracket becomes

$$\begin{aligned}
&\frac{\psi U_c}{\theta f}\,\xi + 1 - \chi_t^{\eta} + \eta\,\frac{\xi}{y} + \eta\left(1-\bar{\tau}_t^y\right)\frac{\xi_c}{c} \\
&\qquad = 1 + (\xi-\gamma_c)\frac{\psi U_c}{\theta f} + \eta\left(1-\bar{\tau}_t^y\right)\frac{\sigma}{c} - \eta\,\frac{\gamma_c}{y} + \eta\,\frac{\xi}{y} + \eta\left(1-\bar{\tau}_t^y\right)\frac{\xi_c}{c} \\
&\qquad = 1 + (\xi-\gamma_c)\frac{\psi U_c}{\theta f} + \eta\,\frac{\xi-\gamma_c}{y} + \eta\left(1-\bar{\tau}_t^y\right)\frac{\sigma+\xi_c}{c} \\
&\qquad = 1 + (\xi-\gamma_c)\left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y}\right) + \eta\left(1-\bar{\tau}_t^y\right)\frac{\sigma+\xi_c}{c}
\end{aligned}$$

so that

$$\mu_t = -\int_0^\infty \left[1 + (\xi-\gamma_c)\left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y}\right) + \eta\left(1-\bar{\tau}_t^y\right)\frac{\sigma+\xi_c}{c}\right]\frac{U_S}{U_c}\,f\,d\theta$$

The envelope iteration that assembles ν_t from the future stock multipliers is that of Appendix B.1 unchanged: the wedge constraints at future dates depend on S only through the utility function, and their effect on continuation costs is already internalized in the future multipliers μ_{t+j}, so $\nu_t = \sum_{j=0}^{s} R^{-j}\mu_{t+j}\Phi_j$. Substituting the damage block and writing the date-$(t+j)$ population integrals as expectations over future skill histories gives the SCC_t of Result 7.2. At $\eta \equiv 0$ the weight collapses to $1 + (\xi - \gamma_c)\,\psi U_c/(\theta f) = 1 + \left(\frac{\tau^y}{1-\tau^y}\right)\frac{\xi-\gamma_c}{1+\epsilon-\gamma_c}$, the fiscal factor of Appendix B.1. The new piece is an income effect: $(\sigma+\xi_c)/c = \partial \ln(U_S/U_c)/\partial c$ is the sensitivity of the marginal damage valuation to the numéraire, and it is priced at $\eta(1-\bar{\tau}^y_t)$ exactly as $\mathcal{I}_t$ is in the commodity wedge. Because it does not vanish under $U = V(h(c,q,S),l)$, the benchmark that made the second-best SCC_t purely Pigouvian in Section 5 no longer does so in the third best unless $\eta_t = 0$.

D.5 The optimal flat schedule

Let the schedule be flat, $\bar{\tau}^y_t(\theta) = \bar{\tau}^y_t$, with the level chosen by the planner. The level enters the date-0 Lagrangian only through the date-t wedge constraints, each contributing

$$\eta f \, \frac{\partial}{\partial \bar{\tau}^y_t}\left[-\frac{U_l}{\theta U_c} - (1-\bar{\tau}^y_t)\right] = \eta f$$

The R^{-t} discounting and the transition densities convert the sum over date-t nodes into the unconditional population expectation, as in Appendix C.9. By the envelope theorem,

$$\frac{dW_0}{d\bar{\tau}^y_t} = \frac{1}{R^t}\,\mathbb{E}_0[\eta_t]$$

so the optimal level sets $\mathbb{E}_0[\eta_t] = 0$. Solving Lemma 7.1 for η_t and taking the expectation,

$$\mathbb{E}_0[\eta_t] = \mathbb{E}_0\left[\frac{1}{\Delta_t}\left(\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t} - A_t\,\frac{\psi U_{c,t}}{\theta f_t}\right)\right] = 0$$

Because the flat rate is common across the cross-section,

$$\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t}\,\mathbb{E}_0\left[\frac{1}{\Delta_t}\right] = \mathbb{E}_0\left[\frac{A_t}{\Delta_t}\,\frac{\psi U_{c,t}}{\theta f_t}\right]$$

The weight $1/\Delta_t$ is a compensated earnings response. Consider the hours problem the frozen schedule decentralizes: the agent chooses l at the net-of-tax rate $\omega \equiv 1 - \bar{\tau}^y_t$, with first-order condition

$$h \equiv \omega\theta\, U_c + U_l = 0$$

Perturb ω holding utility fixed. The compensation is $dc_0 = -\theta l\, d\omega$, so consumption moves only through hours, $dc = \omega\theta\, dl$, and

$$dh = \theta U_c\, d\omega + \left[\omega^2\theta^2 U_{cc} + 2\omega\theta U_{cl} + U_{ll}\right] dl = 0$$

The bracket is the second-order condition of the hours choice, and it is proportional to Δ_t. Substituting $U_{cc} = -\sigma U_c/c$, $U_{cl} = \gamma_c U_c/l$, and $U_{ll} = \epsilon U_l/l = -\epsilon\, \omega\theta U_c/l$, which uses $U_l = -\omega\theta U_c$ at the constrained allocation,

$$\begin{aligned}
\omega^2\theta^2 U_{cc} + 2\omega\theta U_{cl} + U_{ll} &= -\omega\theta U_c \left[\omega\theta\, \frac{\sigma}{c} - \frac{2\gamma_c}{l} + \frac{\epsilon}{l}\right] \\
&= -\omega\theta^2 U_c \left[\omega\, \frac{\sigma}{c} + \frac{\epsilon - 2\gamma_c}{y}\right] = -\omega\theta^2 U_c\, \Delta_t
\end{aligned}$$

with $y = \theta l$, so $\Delta_t > 0$ wherever the hours choice is an interior maximum. Hence,

$$dh = \theta U_c\, d\omega - \omega\theta^2 U_c\, \Delta_t\, dl = 0 \qquad \Longrightarrow \qquad \left.\frac{dl}{d\omega}\right|_u = \frac{1}{\omega\theta\, \Delta_t}$$

With $y = \theta l$,

$$s_{yy,t} \equiv \left.\frac{\partial y}{\partial\,(1 - \bar{\tau}_t^y)}\right|_u = \theta \left.\frac{dl}{d\omega}\right|_u = \frac{1}{(1 - \bar{\tau}_t^y)\, \Delta_t} \qquad \Longrightarrow \qquad \frac{1}{\Delta_t} = (1 - \bar{\tau}_t^y)\, s_{yy,t}$$

Substituting into the optimality condition, with $1 - \bar{\tau}_t^y$ common,

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = \frac{\mathbb{E}_0\left[s_{yy,t}\, A_t\, \frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[s_{yy,t}]}$$

This is Result 7.3. At node-specific rates the same condition without the integral returns $\bar{\tau}^y/(1-\bar{\tau}^y) = A_t \frac{\psi U_c}{\theta f}$, the virtual wedge; the optimal flat rate is its compensated-response-weighted average, exactly as the optimal uniform commodity tax of Appendix C.9 averages the commodity wedges node by node.

E Third-best derivations: the frozen savings wedge

This appendix derives the results of Section 7.2 with every intermediate step shown. Multiplier conventions follow Appendix D; the wedge constraint is now the intertemporal condition (3).

E.1 The problem and first-order conditions

The constraint at date t, node θ^t, requires $\mathbb{E}_t[U_{c,t+1}] = U_{c,t}/((1-\bar{\tau}_t^s(\theta))\,\beta R)$. It involves the next period's controls, so it enters the recursive problem through the marginal-utility promise state of Section 7.2: the date-t problem inherits the requirement (4) with multiplier ϑ_t and passes each successor the state

$$M(\theta) = \frac{U_c\Big(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\Big)}{(1-\bar{\tau}_t^s(\theta))\,\beta R},$$

which is pinned by the current allocation and is not a free control. With the Lagrangian carrying $\vartheta_t\left[\int U_c f\, d\theta - \hat{M}\right]$, the envelope theorem gives $\partial V_t/\partial \hat{M} = -\vartheta_t$. At $t=0$ there is no inherited requirement, so $\vartheta_0 = 0$; at $t = T$ no state is passed, since there is no date $T+1$. The Hamiltonian is

$$\begin{aligned}
\mathcal{H} = &\left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, M(c,q,l,S), \theta, S)\right] f \\
&+ \psi\left[-U_l(c,q,y/\theta,S)\frac{l}{\theta} + \beta w_2\right] \\
&- \lambda_1 u(\theta)\bar{\alpha}_t f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi\left[u - U(c,q,y/\theta,S) - \beta w\right] \\
&- \mu\left[S - S(\ldots, Q)\right] \\
&+ \vartheta_t\, U_c(c,q,y/\theta,S)\, f.
\end{aligned}$$

Differentiating the continuation term with respect to a control $z \in \{c, q, l, S\}$, by the chain rule through M and the envelope condition at $t+1$,

$$\frac{1}{R}\frac{\partial V_{t+1}}{\partial \hat{M}}\frac{\partial M}{\partial z} f = \frac{1}{R}\left(-\vartheta_{t+1}(\theta^t)\right)\frac{U_{cz}}{(1-\bar{\tau}_t^s)\,\beta R} f = -\frac{\vartheta_{t+1}(\theta^t)}{(1-\bar{\tau}_t^s)\,\beta R^2} U_{cz}\, f$$

Adding the $\vartheta_t U_{cz} f$ term from the inherited requirement, every first-order condition gains the combination

$$\left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1-\bar{\tau}_t^s)\,\beta R^2}\right] U_{cz}\, f = \frac{\eta_t^s}{U_{c,t}}\, U_{cz}\, f$$

with η_t^s as defined in Section 7.2. The first-order conditions are as follows:

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: \; f - \psi U_{cl}\frac{l}{\theta} + \frac{\eta_t^s}{U_c} U_{cc}\, f = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: \; f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \frac{\eta_t^s}{U_c} U_{cq}\, f = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: \; -\theta f - \phi U_l + \frac{\eta_t^s}{U_c} U_{cl}\, f = \frac{1}{\theta}\psi\left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \; \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \; \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \frac{\eta_t^s}{U_c} U_{cS}\, f\right) d\theta = \mu,
\end{aligned}$$

with ν_t the shadow price of current emissions of (14). In the statistics of Table 1, $U_{cc}/U_c = -\sigma/c$, $U_{cq}/U_c = -\sigma_{cq}/q$, $U_{cl}/U_c = \gamma_c/l$, and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, so the new term in each condition is η_t^s times the log-derivative of U_c in the corresponding direction. The constraint involves neither u, w, nor w_2, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the envelope identities for $\hat{w}$ and $\hat{w}_2$ are unchanged.

E.2 The multiplier's law of motion

Dividing the first-order condition for c by $U_c f$, using $U_{cl} l = \gamma_c U_c$ and $U_{cc}/U_c = -\sigma/c$,

$$\frac{U_c\,\phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma}{c} \;\equiv\; 1 - \chi_t^s \tag{31}$$

with $\chi_t^s \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s \sigma_t / c_t$. The chain of Appendix A.2 is unchanged, because the constraint enters none of its steps. The first-order condition for w and the envelope condition give

$$\lambda_{1,t+1} = \beta R \frac{\phi}{f} = \beta R \frac{1-\chi_t^s}{U_{c,t}}$$

Integrating the costate equation over $(0,\infty)$ with $\psi(0) = \psi(\infty) = 0$,

$$\lambda_{1,t} = \int_0^\infty \phi \, d\theta = \mathbb{E}_{t-1}\left[\frac{1-\chi_t^s}{U_{c,t}}\right]$$

Evaluating the second expression at $t+1$ and equating with the first,

$$\frac{1-\chi_t^s}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\right] \tag{32}$$

This is the generalized inverse Euler equation of Lemma 7.2. Multiplying (32) by $U_{c,t}$ and substituting the frozen constraint $U_{c,t} = (1-\bar{\tau}_t^s)\beta R\,\mathbb{E}_t[U_{c,t+1}]$,

$$1-\chi_t^s = (1-\bar{\tau}_t^s)\,\mathbb{E}_t[U_{c,t+1}]\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\right]$$

Because $\mathbb{E}[X]\;\mathbb{E}[Y] = \mathbb{E}[XY] - \mathrm{Cov}(X,Y)$, with $X = U_{c,t+1}$ and $Y = (1-\chi_{t+1}^s)/U_{c,t+1}$,

$$1-\chi_t^s = (1-\bar{\tau}_t^s)\left\{\mathbb{E}_t\big[1-\chi_{t+1}^s\big] - \mathrm{Cov}_t\bigg(U_{c,t+1},\,\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\bigg)\right\} \tag{33}$$

When the date-$(t+1)$ draw is degenerate the covariance vanishes and (33) reads

$$1-\chi_t^s = (1-\bar{\tau}_t^s)\left(1-\chi_{t+1}^s\right)$$

Writing $\chi^s = \gamma_c\,\psi U_c/(\theta f) + \eta^s\sigma/c$ on both sides,

$$\left(1-\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t}\right) - \eta_t^s\frac{\sigma_t}{c_t} = (1-\bar{\tau}_t^s)\left[\left(1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}\right) - \eta_{t+1}^s\frac{\sigma_{t+1}}{c_{t+1}}\right]$$

Dividing through by $1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}$, with the virtual savings wedge defined by

$$1-\tau_t^{s,*} \equiv \frac{1-\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t}}{1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}},$$

gives

$$(1-\tau_t^{s,*}) - \frac{\eta_t^s\,\sigma_t/c_t}{1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}} = (1-\bar{\tau}_t^s) - \frac{(1-\bar{\tau}_t^s)\,\eta_{t+1}^s\,\sigma_{t+1}/c_{t+1}}{1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}}$$

Multiplying by $1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}$ and collecting the η^s terms on the left,

$$\eta_t^s\,\frac{\sigma_t}{c_t} - (1-\bar{\tau}_t^s)\,\eta_{t+1}^s\,\frac{\sigma_{t+1}}{c_{t+1}} = \left(1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}\right)(\bar{\tau}_t^s - \tau_t^{s,*}) \tag{34}$$

This is the law of motion of Lemma 7.2. Its boundary conditions come from the definition of η^s: at $t=0$ there is no inherited requirement, so $\vartheta_0 = 0$ and $\eta_0^s = -\vartheta_1(\theta^0)\,U_{c,0}/((1-\bar{\tau}_0^s)\,\beta R^2)$; at $t=T$ no requirement is passed, so $\eta_T^s = \vartheta_T\,U_{c,T}$. The chain $\vartheta_{t+1}(\theta^t) = (1-\bar{\tau}_t^s)\,\beta R^2\,[\vartheta_t - \eta_t^s/U_{c,t}]$ recovers the raw multipliers from the η^s sequence.

Sign reversal. Suppose $\eta^s \geq 0$ at every node; the case $\eta^s \leq 0$ is symmetric. Write $\kappa_t \equiv (1-\bar{\tau}_t^s)\,\beta R^2 > 0$. Along any path: at $t=0$, $\vartheta_0 = 0$ and $\eta_0^s \geq 0$ give $\vartheta_1 \leq 0$; at interior dates, $\eta_t^s \geq 0$ gives $\vartheta_{t+1} \leq \kappa_t\,\vartheta_t$, so $\vartheta_t \leq 0$ propagates forward and $\vartheta_t \leq 0$ for all $t \geq 1$. At the final date, $\eta_T^s = \vartheta_T U_{c,T} \geq 0$ forces $\vartheta_T = 0$. Walking back: $0 = \vartheta_T \leq \kappa_{T-1}\vartheta_{T-1}$ gives $\vartheta_{T-1} \geq 0$, hence $\vartheta_{T-1} = 0$, and by induction $\vartheta_t = 0$ at every node, so $\eta^s \equiv 0$. A nonzero multiplier profile therefore takes both signs, which is Corollary 7.2.

The two-period example. Let $T=1$, preferences separable ($\gamma_c = \sigma_{cq} = 0$, so $\tau_0^{s,*} = 0$), and $\bar{\tau}_0^s > 0$. The boundary conditions give $\eta_0^s = -\vartheta_1 U_{c,0}/\kappa_0$ and $\eta_1^s = \vartheta_1 U_{c,1}$. At $\gamma_c = 0$ the right-hand side of (34) at $t=0$ is $\bar{\tau}_0^s - \tau_0^{s,*} = \bar{\tau}_0^s$; substituting both boundary expressions into the left,

$$-\vartheta_1\left[\frac{U_{c,0}\,\sigma_0}{\kappa_0\,c_0} + (1-\bar{\tau}_0^s)\,\frac{U_{c,1}\,\sigma_1}{c_1}\right] = \bar{\tau}_0^s \qquad \Longrightarrow \qquad \vartheta_1 = -\frac{\bar{\tau}_0^s}{\dfrac{U_{c,0}\,\sigma_0}{\kappa_0\,c_0} + (1-\bar{\tau}_0^s)\,\dfrac{U_{c,1}\,\sigma_1}{c_1}} \;<\; 0,$$

so $\eta_0^s > 0 > \eta_1^s$: with savings overtaxed, the carbon price sits above the social cost of carbon at date 0 and below it at date 1.

E.3 Labor wedge

From the first-order condition for l, using $U_{ll}l + U_l = U_l(1+\epsilon)$,

$$\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \frac{\eta_t^s}{U_c}\frac{U_{cl}}{U_l}f$$

Now $U_l = -(1-\tau_t^y)\theta U_c$, where τ_t^y is the endogenous labor wedge, so $-\theta f/U_l = f/((1-\tau_t^y)U_c)$, and with $U_{cl} = \gamma_c U_c/l$,

$$\frac{\eta_t^s}{U_c}\frac{U_{cl}}{U_l} = \frac{\eta_t^s}{U_c}\frac{\gamma_c U_c/l}{-(1-\tau_t^y)\theta U_c} = -\frac{\eta_t^s\,\gamma_c}{(1-\tau_t^y)\,y\,U_c}$$

Multiplying by $\frac{U_c}{f}$,

$$\frac{U_c\,\phi}{f} = \frac{1}{1-\tau_t^y} - (1+\epsilon)\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\gamma_c}{(1-\tau_t^y)\,y} \tag{35}$$

Equating (31) and (35),

$$1 - \gamma_c\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma}{c} = \frac{1}{1-\tau_t^y} - (1+\epsilon)\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\gamma_c}{(1-\tau_t^y)\,y}$$

Moving 1 to the right and the $\psi U_c/(\theta f)$ and η_t^s terms to the left, with $\frac{1}{1-\tau_t^y} - 1 = \frac{\tau_t^y}{1-\tau_t^y}$ and $A_t = 1+\epsilon-\gamma_c$,

$$\frac{\tau_t^y}{1-\tau_t^y} = (1+\epsilon-\gamma_c)\frac{\psi U_c}{\theta f} - \eta_t^s\left[\frac{\sigma}{c} - \frac{\gamma_c}{(1-\tau_t^y)\,y}\right] = A_t\frac{\psi U_c}{\theta f} - \eta_t^s\left[\frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{(1-\tau_t^y)\,y_t}\right]$$

This is the first formula of Result 7.4. The bracket is minus the log-derivative of U_c along the utility-compensated earning direction $(dc, dl) = (1-\tau_t^y,\ 1/\theta)$, expressed per net-of-tax dollar of output, the intratemporal image of the delivery-drag statistics of Appendix D. At $\eta_t^s = 0$ the condition is (7).

E.4 Commodity wedge

Dividing the first-order condition for q by U_q, using $U_{ql}l = \gamma_q U_q$,

$$\phi = \frac{(1+\nu_t) f}{U_q} - \frac{\psi}{\theta}\gamma_q + \frac{\eta_t^s}{U_c}\frac{U_{cq}}{U_q}\, f$$

Multiplying by $\frac{U_c}{f}$, with $U_c/U_q = 1/(1+\tau^q)$ and $U_{cq}/U_c = -\sigma_{cq}/q$,

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{1+\tau^q} - \gamma_q\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} \tag{36}$$

Equating (36) with (31),

$$1 - \gamma_c\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma}{c} = \frac{1+\nu_t}{1+\tau^q} - \gamma_q\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma_{cq}}{(1+\tau^q)\,q}$$

After collecting terms, with $1 - \frac{1+\nu_t}{1+\tau^q} = \frac{\tau^q-\nu_t}{1+\tau^q}$,

$$\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s\left[\frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{(1+\tau_t^q)\,q_t}\right] = (\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s\,\mathcal{I}_t$$

This is the second formula of Result 7.4, with $\mathrm{SCC}_t \equiv \nu_t$ and the same income-effect gap $\mathcal{I}_t$ as in Appendix C.3, whose normality reading applies verbatim.

E.5 The costate

Rearranging (31),

$$\phi = \frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c - \eta_t^s\,\frac{\sigma}{c}\,\frac{f}{U_c}$$

Substituting into the costate equation $\dot{\psi} = \lambda_1\bar{\alpha}_t f - \lambda_2 f_2 - \phi$ and moving the γ_c term to the left,

$$\dot{\psi} - \frac{\gamma_c}{\theta}\psi = \lambda_1\bar{\alpha}_t f - \lambda_2 f_2 - \left[1 - \eta_t^s\,\frac{\sigma}{c}\right]\frac{f}{U_c}$$

Unlike Appendix D.3, the multiplier η_t^s does not depend on the costate, so the integrating factor is that of Appendix A.1 unchanged, and only the source is reweighted. Multiplying by $\exp\left[-\int^{\theta}\gamma_c(\tilde{x})\,d\tilde{x}/\tilde{x}\right]$ and integrating from θ to ∞ with $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] \left(\left[1 - \eta_t^s(x)\, \frac{\sigma(x)}{c(x)}\right] \frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

with $\psi(0) = 0$ pinning $\lambda_{1,t}$ as in (9). Carrying $U_c(\theta)/U_c(x)$ into the integrand as in identity (10), the product $A_t \frac{\psi U_c}{\theta f}$ inherits the $B_t C_t D_t$ structure of Appendix A.1 with the redistributive weight $1 - \eta_t^s(x)\sigma(x)/c(x) - \lambda_{1,t}\bar{\alpha}_t(x) U_{c,t}(x)$ inside C_t: the resource value of taking from type x is corrected by the constraint's valuation of that type's marginal utility. The threat-keeping multiplier is unchanged in form, $\lambda_{2,t} = \beta R\, \psi_{t-1}/f_{t-1}$, with $\psi_{t-1} U_{c,t-1}/(\theta_{t-1} f_{t-1})$ read from the labor-wedge formula of Appendix E.3.

E.6 The social cost of carbon

The first-order condition for S determines the date-t stock multiplier,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S + \frac{\eta_t^s}{U_c}\, U_{cS}\, f \right) d\theta$$

Each piece can be expressed in the marginal damage valuation U_S/U_c. Using $U_{Sl} l = \xi U_S$,

$$-\psi U_{Sl} \frac{l}{\theta} = -\frac{\psi}{\theta}\, \xi\, U_S = -\frac{\psi U_c}{\theta f}\, \xi\, \frac{U_S}{U_c}\, f$$

From (31),

$$-\phi U_S = -\left(1 - \chi_t^s\right) \frac{U_S}{U_c}\, f$$

With $U_{cS}/U_c = \xi_c\, (U_S/U_c)/c$,

$$\frac{\eta_t^s}{U_c}\, U_{cS}\, f = \eta_t^s\, \frac{\xi_c}{c}\, \frac{U_S}{U_c}\, f$$

Hence,

$$\mu_t = -\int_0^\infty \left[\frac{\psi U_c}{\theta f}\, \xi + 1 - \chi_t^s - \eta_t^s\, \frac{\xi_c}{c} \right] \frac{U_S}{U_c}\, f\, d\theta$$

Expanding $1-\chi_t^s = 1-\gamma_c\frac{\psi U_c}{\theta f} - \eta_t^s\sigma/c$ and collecting,

$$\frac{\psi U_c}{\theta f}\,\xi + 1 - \gamma_c\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma}{c} - \eta_t^s\,\frac{\xi_c}{c} = 1 + (\xi-\gamma_c)\,\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma+\xi_c}{c}$$

so that

$$\mu_t = -\int_0^\infty \left[1 + (\xi-\gamma_c)\,\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma+\xi_c}{c}\right]\frac{U_S}{U_c}\,f\,d\theta$$

The envelope iteration $\nu_t = \sum_{j=0}^{s} R^{-j}\mu_{t+j}\Phi_j$ is unchanged, giving the damage weight of Result 7.4. The income-effect piece is the same $\partial\ln(U_S/U_c)/\partial c = (\sigma+\xi_c)/c$ as in Appendix D.4, now priced at $-\eta_t^s$; at $\eta^s \equiv 0$ the weight collapses to the fiscal factor of Appendix B.1.

E.7 The optimal flat rate

Let the schedule be flat, $\bar\tau_t^s(\theta) = \bar\tau_t^s$, with the level chosen by the planner. The rate enters the date-0 Lagrangian only through the states $M(\theta)$ passed forward at date t (Appendix E.1). By the chain rule and the envelope condition $\partial V_{t+1}/\partial\hat M = -\vartheta_{t+1}$, each date-$t$ node contributes

$$\frac{1}{R}\,\frac{\partial V_{t+1}}{\partial \hat M}\,\frac{\partial M}{\partial\bar\tau_t^s}\,f = \frac{1}{R}\,(-\vartheta_{t+1})\,\frac{U_{c,t}}{(1-\bar\tau_t^s)^2\,\beta R}\,f = -\,\frac{\vartheta_{t+1}\,U_{c,t}}{(1-\bar\tau_t^s)^2\,\beta R^2}\,f$$

Aggregating over date-t nodes as in Appendix D.5, with the common factor $(1-\bar\tau_t^s)^{-2}\,\beta^{-1}R^{-2}$ pulled out, the envelope theorem gives

$$\frac{dW_0}{d\bar\tau_t^s} = -\,\frac{R^{-t}}{(1-\bar\tau_t^s)^2\,\beta R^2}\;\mathbb{E}_0[\vartheta_{t+1}\,U_{c,t}]$$

so the optimal level sets $\mathbb{E}_0[\vartheta_{t+1}U_{c,t}] = 0$. For the equivalence in Result 7.5, rearranging the definition of the composite multiplier, $\eta_t^s = \vartheta_t U_{c,t} - \vartheta_{t+1}U_{c,t}/((1-\bar\tau_t^s)\,\beta R^2)$,

$$\frac{\vartheta_{t+1}\,U_{c,t}}{(1-\bar\tau_t^s)\,\beta R^2} = \vartheta_t\,U_{c,t} - \eta_t^s$$

Taking $\mathbb{E}_0$ of both sides, the optimality condition reads $\mathbb{E}_0[\eta_t^s] = \mathbb{E}_0[\vartheta_t U_{c,t}]$; at $t=0$, $\vartheta_0 = 0$ gives $\mathbb{E}_0[\eta_0^s] = 0$. Only the forward component of the multiplier is priced, because the inherited requirements at date t were set by the date-$(t-1)$ rate. Optimization therefore disciplines the multiplier's weighted averages, not its profile: η^s generally remains nonzero node by node, the sign-reversal argument of Appendix E.2 is unaffected, and the carbon-price tilt of Corollary 7.2 survives at the optimal flat path.